
High-Accuracy Sampling for Diffusion Models and Log-Concave Distributions

Fan Chen¹ Sinho Chewi² Constantinos Daskalakis¹ Alexander Rakhlin¹

Abstract

We present algorithms for diffusion model sampling which obtain δ -error in $\text{polylog}(1/\delta)$ steps, given access to $\tilde{O}(\delta)$ -accurate score estimates in L^2 . This is an exponential improvement over all previous results. Specifically, under minimal data assumptions, the complexity is $\tilde{O}(d \text{polylog}(1/\delta))$ where d is the dimension of the data; under a non-uniform L -Lipschitz condition, the complexity is $\tilde{O}(\sqrt{dL} \text{polylog}(1/\delta))$; and if the data distribution has intrinsic dimension d_* , then the complexity reduces to $\tilde{O}(d_* \text{polylog}(1/\delta))$. Our approach also yields the first $\text{polylog}(1/\delta)$ complexity sampler for general log-concave distributions using only gradient evaluations.¹

1. Introduction

What is the complexity of sampling from a continuous probability distribution, given access to evaluations of the gradient of the log-density? In particular, can one design algorithms whose iteration complexity scales as $\text{polylog}(1/\delta)$, where δ is the target accuracy, or must they necessarily take $\text{poly}(1/\delta)$ steps?

Complexity bounds which scale as $\text{polylog}(1/\delta)$, indicating that algorithms converge exponentially fast, are known as “high-accuracy” guarantees, as they ensure that one can draw an extremely accurate sample without too many iterations. When access to the unnormalized log-density of the target distribution is available, and not just its gradient, then high-accuracy samplers abound, based on accept-reject mechanisms such as rejection sampling or the Metropolis–Hastings filter.

Without density evaluations, the answer is less clear. In this setting, existing sampling methods are typically based

on discretizations of stochastic differential equations, and the need to control the discretization error precludes high-accuracy guarantees. (Note that this is unlike the setting of optimization, in which gradient descent enjoys a high-accuracy guarantee under strong convexity—there, discretization does not bias the algorithm.) A notable exception is discretization of piecewise deterministic Markov processes (PDMPs), for which Lu & Wang (2022) show that $\text{polylog}(1/\delta)$ evaluations of the gradient suffice from a warm start.

This question has become particularly interesting in light of recent developments on diffusion-based generative modeling. Such models are based on implementing a certain reverse Markov process, in which each iteration requires evaluation of a score function—the gradient of the log-density along a diffusion process. Since these models only learn the score function and not the (unnormalized) density function itself (unlike earlier approaches such as energy-based models), it is natural to ask what the best achievable complexity is using score evaluations alone.

More precisely, we are interested in bounding the number of steps (and queries to the score function) required to sample from a distribution $\hat{p}$ such that

$$D(p_{\text{data}}, \hat{p}) \leq \delta + C_{\text{apx}} \cdot \varepsilon_{\text{score}}, \quad (1)$$

where D is a measure of discrepancy, $\delta \in (0, 1)$ is the target accuracy, $\varepsilon_{\text{score}}$ measures the error in the score function estimates, and C_{apx} is a target approximation factor.

The initial work of Chen et al. (2023c); Lee et al. (2023) showed that with L^2 -accurate score estimates, Denoising Diffusion Probabilistic Models (DDPMs) achieve a query complexity of $1/\delta^2$ in total variation distance for sampling from an early stopped distribution, provided that p_{data} has bounded support, with an overhead that is polynomial in the dimension and radius, and with $C_{\text{apx}} = \tilde{O}(1)$. By a standard argument, this can be converted to guarantees for sampling from p_{data} itself in a weaker metric. Importantly, these results impose almost no assumptions on the data distribution, providing strong theoretical justification for diffusion models.

Since then, there has been an explosion of works aimed at extending and refining these guarantees (e.g., Chen et al., 2023a; Li et al., 2023; Benton et al., 2024; Gao et al., 2025;

¹Massachusetts Institute of Technology, Massachusetts, USA
²Yale University, Connecticut, USA. Correspondence to: Fan Chen <fanchen@mit.edu>.

Proceedings of the 43rd International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026. Copyright 2026 by the author(s).

¹Since the initial submission of this work, we have obtained improved results; details are available in the [arxiv version](#) of this paper.

Conforti et al., 2025; Li & Yan, 2025; Chen et al., 2023b; Li et al., 2024b; Gao & Zhu, 2025; Huang et al., 2025a; Jain & Zhang, 2026), which we cannot fully survey here. However, it is known that a complexity of $\Omega(1/\delta)$ is unimprovable for DDPM (Jiao et al., 2025), which motivates the development of alternative algorithms. Under minimal assumptions, the algorithm of Li & Cai (2024) achieves a query complexity of $1/\delta^{1/2}$. Other works achieved similar speed-ups assuming that the Jacobian of the score error is bounded (Li et al., 2024a; 2025b).

A recent line of works (e.g., Huang et al., 2025a;b; Li et al., 2025b) studied higher-order discretization methods, with the resulting query complexity scaling roughly as $C_P d^{1+1/p}/\delta^{1/p}$, where $p \geq 1$ is the chosen acceleration order, and C_P hides dependence on other problem parameters² along with an implicit (often exponential) dependence on p . These results show that diffusion sampling can be performed with sub-polynomial dependence on $1/\delta$,³ but they still fall well short of the desired poly-logarithmic complexity.⁴

Meanwhile, Huang et al. (2024a); Wainwright (2025) propose to use unnormalized density evaluations to achieve high-accuracy samplers, and Huang et al. (2025c) propose a high-accuracy sampler which requires learning a certain “quantized” score. As discussed above, these are not compatible with the current practice of diffusion sampling, which only learns score estimates.

Therefore, we state our main question of interest:

Is there a diffusion model sampler which achieves a high-accuracy guarantee using only score evaluations, under minimal assumptions on both the data distribution and the score error?

1.1. Our Contribution

We answer this question affirmatively via a new meta-algorithm, which we call *first-order rejection sampling* (FORS); see Algorithm 1. This algorithm aims at simulating rejection sampling using only first-order (gradient) queries. We give consequences of our method for diffusion sampling and for log-concave sampling.

²For example, in the results of Huang et al. (2024a; 2025a;b), C_P hides polynomial dependence on higher-order Lipschitz parameters or the diameter of the support of p_{data} .

³Roughly, we should expect $C_P \geq e^{\Omega(p)}$. Hence, the complexity is at least $d \exp(O(\sqrt{\log(d/\delta)}))$ with the best choice of acceleration order p , which is better than $\text{poly}(1/\delta)$ but much worse than $\text{poly}(\log(1/\delta))$.

⁴Furthermore, the approximation factor C_{apx} in Huang et al. (2025a;b) scales polynomially in the early stopping time, and Li et al. (2025b) additionally relies on the second-order score error $\varepsilon_{\text{Jacobi}}$, making the results sensitive to score errors.

Diffusion Sampling. We show the following results, where the error is measured in the bounded Lipschitz metric and $\varepsilon_{\text{score}}$ denotes the L^2 -error of the score estimates (Definition 2.1).

- Under **minimal data assumptions**—namely, p_{data} has a finite second moment M_2^2 —we obtain δ error in $O(d \log^2(M/\delta) + \log^3(M/\delta))$ queries with $C_{\text{apx}} = O(1)$, where $M := d + M_2^2$.

This result strictly improves upon all prior results in the literature in this setting.

Beyond this setting, many works in the literature aim to sample with a number of steps which is sublinear in the dimension (Chen et al., 2023b; Jiao & Li, 2024; Jiao et al., 2025; Zhang et al., 2025). In an orthogonal direction, other works aim to show that the number of steps depend on the *intrinsic* dimension of the data (Azangulov et al., 2024; Huang et al., 2024b; Li & Yan, 2024; Potapchik et al., 2024; Li et al., 2025a; Liang et al., 2025; Tang & Yan, 2025). We also incorporate these advances into our framework.

- Assuming **Lipschitz score functions**—where we use the significantly weaker notion of a “non-uniform” Lipschitz constant L (Assumption 4.3)—we obtain δ error in $O(\sqrt{dL} \log^{3/2}(M/\delta) + L \log^2(M/\delta))$ steps, where L is the Lipschitz constant, and with $C_{\text{apx}} = O(\sqrt{d/L})$.
- For **low-dimensional data distributions**, we obtain δ error in $O(d_* \log^2(M/\delta) + \log^3(M/\delta))$ steps, where d_* is the intrinsic dimension (Definition 4.6), and with $C_{\text{apx}} = O(d/d_*)$ under a slightly stronger notion of error in the score estimates.

Log-Concave Sampling. For sampling from log-concave (and isoperimetric) densities with gradient evaluations of the log-density, we recover state-of-the-art results from Fan et al. (2023), except that *we do not require density evaluations*; see Section 5.

1.2. Related Work

As we do not have space to survey the vast literature on diffusion model guarantees, we focus on the ones most relevant to our work.

High-Accuracy Diffusion Sampling. Prior to this work, Huang et al. (2024a); Wainwright (2025) studied diffusion sampling with additional *zeroth-order* queries. Specifically, given access to estimates of the unnormalized log-densities with bounded error, query complexity bounds of $\tilde{O}(d^2 \log(1/\delta))$ (Huang et al., 2024a) and $\tilde{O}(\sqrt{d} \log^3(1/\delta))$ (Wainwright, 2025) were shown under additional Lipschitz conditions on the score functions. Both works resort to standard high-accuracy sampling methods

(e.g., Metropolis-adjusted Langevin). However, estimating the densities can be extremely challenging in practice. Moreover, [Huang et al. \(2025c\)](#) propose a method based on learning a quantized score, which requires changing the score matching objective.

Concurrent Work. In concurrent work, [Gatmiry et al. \(2026\)](#) also obtains a high-accuracy guarantee for diffusion sampling. Assuming that p_{data} is a Gaussian convolution, i.e., $p_{\text{data}} = p_{\star} * \mathcal{N}(0, \sigma^2 I)$ where $p_{\star}$ is supported on a ball of radius R , and that the score errors are bounded in the *sub-exponential* norm, they obtain a query complexity of $\tilde{O}((R/\sigma)^2 \log^2(1/\delta))$. While their assumptions are stronger than ours, the final complexity bounds are incomparable. We defer a lengthier comparison to [Appendix A.1](#).

In addition, the concurrent work of [Xun & Price \(2026\)](#) establishes a query complexity lower bound of $\tilde{\Omega}(\sqrt{d})$ for diffusion sampling when p_{data} is a d -dimensional distribution with bounded support. Together with our $\tilde{O}(d)$ upper bound in this setting, this leaves a gap of at most $\tilde{O}(\sqrt{d})$; closing it is an interesting direction for future work.

Log-Concave Sampling. Our application to log-concave sampling is based on the proximal sampler ([Lee et al., 2021](#); [Chen et al., 2022](#)), and our results can be compared to [Altschuler & Chewi \(2024\)](#); [Fan et al. \(2023\)](#). See the book draft [Chewi \(2026\)](#) for an overview of the subject.

1.3. Notation

We make use of the following divergences between probability distributions: the *total variation* (TV) distance $D_{\text{TV}}(\mu, \nu) := \frac{1}{2} \int |\mu - \nu|$; the *Hellinger distance* $D_{\text{H}}^2(\mu, \nu) := \frac{1}{2} \int (\sqrt{\mu} - \sqrt{\nu})^2$; the *KL divergence* $D_{\text{KL}}(\mu \parallel \nu) := \mathbb{E}_{\mu} \log(\mu/\nu)$; the *chi-squared divergence* $D_{\chi^2}(\mu \parallel \nu) := \mathbb{E}_{\mu}(\mu/\nu)^2 - 1$; and the *Wasserstein distance* $W_2^2(\mu, \nu) := \inf_{\gamma} \text{coupling of } \mu, \nu \mathbb{E}_{(X,Y) \sim \gamma} \|X - Y\|^2$. In addition, we consider the bounded Lipschitz metric which metrizes weak convergence:

$$D_{\text{BL}}(\mu, \nu) := \sup\{\mathbb{E}_{\mu} f - \mathbb{E}_{\nu} f : |f| \leq 1, \|f\|_{\text{Lip}} \leq 1\}.$$

We define

$$\text{Clip}_B(x) := \max\{-B, \min\{B, x\}\}, \quad \forall x \in \mathbb{R}.$$

We use $\lesssim$ and $O(\cdot)$ to hide absolute constants, i.e., $f \lesssim g$ (and $f = O(g)$) if there is an absolute constant such that $f \leq Cg$. The notation $\tilde{O}(\cdot)$ hides logarithmic factors.

2. Background on Diffusion Models

Recall that Denoising Diffusion Probabilistic Models (DDPMs) are based on the following forward process that transforms a sample $X_0 \sim p_{\text{data}}$ to noise:

$$X_0 \sim p_{\text{data}}, \quad X_{t+1} \sim \mathcal{N}(\alpha_t X_t, \alpha_t^2 \eta_t I), \quad t \in [T]. \quad (2)$$

Note that this is a Markov chain, and it is easy to see that $X_t \mid X_0 \sim \mathcal{N}(\bar{\alpha}_t X_0, \sigma_t^2 I)$, where

$$\bar{\alpha}_t = \prod_{s=0}^{t-1} \alpha_s, \quad \sigma_t^2 = \alpha_{t-1}^2 (\sigma_{t-1}^2 + \eta_{t-1}).$$

Conversely, given the parameter sequence $(\bar{\alpha}_t, \sigma_t)_{t \in [T]}$, the corresponding $(\alpha_t, \eta_t)_{t \in [T]}$ is given by

$$\alpha_t = \frac{\bar{\alpha}_{t+1}}{\bar{\alpha}_t}, \quad \eta_t = \frac{\sigma_{t+1}^2}{\alpha_t^2} - \sigma_t^2 = \frac{\sigma_{t+1}^2}{\bar{\alpha}_{t+1}^2} \bar{\alpha}_t^2 - \sigma_t^2.$$

We say that the DDPM is *variance-preserving* if $\bar{\alpha}_t^2 + \sigma_t^2 = 1$, and hence $\alpha_t^2 = \frac{1 - \sigma_{t+1}^2}{1 - \sigma_t^2}$, $\sigma_{t+1}^2 = \frac{1 - \sigma_{t+1}^2}{1 - \sigma_t^2} (\sigma_t^2 + \eta_t)$.

We let p_t be the probability density function of X_t , and let $\rho_t(\cdot \mid x')$ be the probability density function of the backward transition kernel $\mathbb{P}(X_t = \cdot \mid X_{t+1} = x')$. Then, by Bayes rule, it holds that

$$\rho_t(x \mid x') \propto_x p_t(x) \exp\left(-\frac{\|x - \alpha_t^{-1} x'\|^2}{2\eta_t}\right). \quad (3)$$

In this work, we focus on the task of sampling from p_1 with extremely small values of $1 - \alpha_0$ and σ_1 . This corresponds to *early stopping*; see [Section 4.1](#) for further discussion.

Score Function and Estimates. As above, we assume that we have access to approximate score functions $(s_t)_{t \in [T]}$ such that $s_t \approx s_t^* := \nabla \log p_t$ with controlled mean-squared error. Recall that by Tweedie's identity,

$$s_t^*(x) = \nabla \log p_t = \frac{1}{\sigma_t^2} \mathbb{E}[\bar{\alpha}_t X_0 - X_t \mid X_t = x], \quad (4)$$

where the conditional expectation is taken over $X_0 \sim p_{\text{data}}$, $X_t \sim \mathcal{N}(\bar{\alpha}_t X_0, \sigma_t^2)$. We define $\tilde{s}_t^*(x) := s_t^*(x) + \frac{x}{\sigma_t^2} = \frac{\bar{\alpha}_t}{\sigma_t^2} \mathbb{E}[X_0 \mid X_t = x]$ to be the rescaled posterior mean, and we also define $\tilde{s}_t(x) := s_t(x) + \frac{x}{\sigma_t^2}$.

Definition 2.1 (Score estimation error). Given the score estimates $(s_t)_{t \in [T]}$, we define

$$\varepsilon_{t, \text{score}}^2 := \mathbb{E}_{X_t \sim p_t} \|s_t(X_t) - s_t^*(X_t)\|^2, \quad t \in [T].$$

Roadmap. To approximately generate a sample from p_{data} , we learn approximate score functions $s_t \approx \nabla \log p_t$. DDPM uses the approximation $\log p_t(x) \approx \log p_t(x') + \langle s_t^*(x'), x - x' \rangle$, yielding

$$\rho_t(\cdot \mid x') \approx \mathcal{N}(\alpha_t^{-1} x' + \eta_t s_t(x'), \eta_t I). \quad (5)$$

In our work, we develop algorithms for sampling from (3) directly. In [Section 3](#), we motivate our approach by assuming that $s_t = \nabla \log p_t$ is *exact*, which has implications for log-concave sampling described in [Section 5](#). In [Section 4](#), we instantiate our methods on diffusion sampling and show that the convergence guarantees are *robust* with respect to the error in the score function estimates.

3. Key Subroutine: Gaussian Tilts

In this section, we focus on the problem of sampling from a Gaussian tilt of the form

$$\nu(x) \propto \exp\left(-f(x) - \frac{\|x - x_0\|^2}{2\eta}\right), \quad (6)$$

where we assume access to first-order queries for f . For our eventual application to diffusion models, we will choose $f = -\log p_t$ in view of Eq. (3) and approximate $-\nabla f \approx s_t$.

3.1. First-Order Rejection Sampling (FORS)

We motivate our approach by considering the simple problem of sampling from a density $p \propto e^{-f}$, where $f : [0, 1] \rightarrow \mathbb{R}$, $f(0) = 0$, and $-1 \leq f' \leq 1$. Our goal is to develop a *high-accuracy sampler*—a sampler whose sampled distribution has δ error in total variation distance to p —in $\text{polylog}(1/\delta)$ steps, using only queries to f' .

Consider performing rejection sampling with the base measure $\text{Unif}([0, 1])$. To do so, we must generate randomness $b \sim \text{Ber}(ce^{-f(x)})$ for any given $x \in [0, 1]$. However, we only have access to f' . The identity $f(x) = \int_0^x f'(y) dy$, written as

$$f(x) = \mathbb{E}_{y \sim \text{Unif}([0, x])}[xf'(y)], \quad (7)$$

suggests an unbiased estimate of $f(x)$. Is it possible to sample from $\text{Ber}(ce^{-f(x)})$ if we have access to an unbiased estimate of $f(x)$? A more general version of this idea is known as the ‘‘Bernoulli factory’’ problem (Keane & O’Brien, 1994; Nacu & Peres, 2005). It can be stated as the following abstract task:

Task: Given i.i.d. random variables $W_1, W_2, W_3, \dots$ in $[-1, 1]$, generate a sample $b \sim \text{Ber}(ce^{\mathbb{E} W_1})$.

To solve this, we write the Taylor series as

$$e^{\mathbb{E} W_1} = e^{-1} \cdot e^{\mathbb{E}[1+W_1]} = \sum_{j \geq 0} \frac{e^{-1}}{j!} (\mathbb{E}[1+W_1])^j.$$

Suppose that $J \sim \text{Poisson}(2)$ is independent of the i.i.d. sequence $W_1, W_2, W_3, \dots$. Then we notice that

$$e^{\mathbb{E} W_1} = e \mathbb{E} \left[\prod_{j=1}^J \left(\frac{1+W_j}{2} \right) \right],$$

and simply set $b \sim \text{Ber}(\prod_{j=1}^J (\frac{1+W_j}{2}))$. Indeed, $\mathbb{P}(b = 1) = \mathbb{E} \prod_{j=1}^J (\frac{1+W_j}{2}) = e^{-1+\mathbb{E} W_1}$.

In summary, we can generate a sample $b \sim \text{Ber}(ce^{-f(x)})$ without having to compute or accurately approximate $f(x)$ (this requires the integration step $f(x) = \int_0^x f'(y) dy$ and can be expensive). Instead, it is sufficient to have access to (a random number of) unbiased estimates of $f(x)$. The

latter can be achieved with derivative information, in view of (7).

We now generalize this setup via the following meta-algorithm, called *first-order rejection sampling (FORS)*. Given a proposal distribution q and a tilt function w , the goal of Algorithm 1 is to produce a sample from $\hat{p}(x) \propto q(x) e^{w(x)}$ without having access to the value $w(x)$. Instead, for each $x \in \mathbb{R}^d$, we can generate i.i.d. samples $W_1, W_2, W_3, \dots$ such that $\mathbb{E}[W_1 | x] = w(x)$. Let $\mathcal{W}_x$ denote the conditional distribution of W_1 given x .

Algorithm 1 First-order rejection sampling (FORS)

Input: Parameter $B > 0$, proposal distribution q over $\mathbb{R}^d$, estimator distributions $(\mathcal{W}_x)_{x \in \mathbb{R}^d}$ supported on $[-B, B]$
for $i = 1, 2, 3, \dots$ **do**
 Sample $x \sim q$.
 Sample $J \sim \text{Poisson}(2B)$.
 Sample i.i.d. $W_1, \dots, W_J \sim \mathcal{W}_x$.
 Output x with probability $\prod_{j=1}^J \frac{B+W_j}{2B}$.
end for

Theorem 3.1 (FORS guarantee). *Algorithm 1 outputs a random point with density $\hat{p}(x) \propto q(x) e^{\mathbb{E}[W_1|x]}$. The number of sampled W_j ’s is bounded, with probability at least $1 - \delta$, by $3Be^{2B} \log(2/\delta)$.*

Moreover, if Algorithm 1 is called T times, then with probability at least $1 - \delta$, the total number of sampled W_j ’s is $O(Be^{2B} (T + \log(1/\delta)))$.

Variants of this idea have also been applied to *exactly* simulate SDEs (e.g., Wagner, 1988; Beskos & Roberts, 2005; Beskos et al., 2006; Papaspiliopoulos, 2011).

3.2. Sampling from Gaussian Tilts with an Exact Oracle

Now, we return to the problem of sampling from a general Gaussian tilt:

$$\nu(x) \propto \exp\left(-f(x) - \frac{\|x - x_0\|^2}{2\eta}\right).$$

By Section 3.1, it suffices to construct a proposal distribution q and a tilt function w such that (a) $\nu(x) \propto q(x) \cdot e^{w(x)}$, and (b) a bounded estimator for $w(x)$ can be constructed from ∇f .

Condition (a) is equivalent to

$$w(x) = -f(x) - \frac{\|x - x_0\|^2}{2\eta} - \log q(x) + \text{const.}$$

To satisfy both (a) and (b), it is natural to choose q as a Gaussian approximation to ν obtained via a first-order expansion of f . Concretely, for a fixed $x_+ \in \mathbb{R}^d$, Taylor expansion gives

$$f(x) \approx f(x_+) + \langle x - x_+, \nabla f(x_+) \rangle,$$

and hence we choose $q = N(x_0 - \eta \nabla f(x_+), \eta I)$ so that

$$q(x) \propto \exp\left(-\langle x - x_+, \nabla f(x_+) \rangle - \frac{1}{2\eta} \|x - x_0\|^2\right).$$

Then, by the construction in [Section 3.1](#), we can express

$$\begin{aligned} & \log \nu(x) - \log q(x) - \text{const} \\ &= \langle x - x_+, \nabla f(x_+) \rangle - f(x) + f(x_+) \\ &= \int_0^1 \langle x - x_+, \nabla f(x_+) - \nabla f(rx + (1-r)x_+) \rangle dr. \end{aligned}$$

In particular, we can set

$$W_{r,x} := \langle x - x_+, \nabla f(x_+) - \nabla f(rx + (1-r)x_+) \rangle,$$

so that

$$\nu(x) \propto q(x) \exp(\mathbb{E}_r W_{r,x}),$$

where the expectation $\mathbb{E}_r[\cdot]$ is taken over $r \sim \text{Unif}([0, 1])$.

Further, assuming that ∇f is β -Lipschitz, we can bound $|W_{r,x}| \leq \beta \|x - x_+\|^2$. Therefore, as long as $x_+ \approx x_0 - \eta \nabla f(x_+)$ and $\beta d \eta \ll 1$, we can guarantee that $\sup_r |W_{r,x}| \leq 1$ with high probability over $x \sim q$. This implies the following distribution is a good approximation of ν :

$$\widehat{\nu}(x) \propto q(x) \exp(\mathbb{E}_r \widehat{W}_{r,x}),$$

where $\widehat{W}_{r,x} = \text{Clip}_B(W_{r,x})$ and $B = \Theta(1)$ is a tuneable parameter.

General Path Integral. We can generalize the above pipeline with any *path integral* and demonstrate (a) that in the gradient Lipschitz case, we in fact only need to ensure $\beta \sqrt{d} \eta \ll 1$, and (b) the Lipschitz continuity of ∇f can be weakened to Hölder continuity for any exponent $s \in [0, 1]$.

Fix a distribution P (to be determined later) over $\mathbb{R}^d$ and a *path function* $\gamma_{z,r}(x) := \gamma(x; z, r)$ such that $\gamma_{z,1}(x) = x$ and $\gamma_{z,0}(x) = \bar{\gamma}(z)$ is independent of x . Then we can express any smooth function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$\begin{aligned} & h(x) - \mathbb{E}_{z \sim P}[h(\bar{\gamma}(z))] \\ &= \int_0^1 \mathbb{E}_{z \sim P} \langle \dot{\gamma}_{z,r}(x), \nabla h(\gamma_{z,r}(x)) \rangle dr \\ &= \mathbb{E}_{z \sim P, r \sim \text{Unif}([0, 1])} \langle \dot{\gamma}_{z,r}(x), \nabla h(\gamma_{z,r}(x)) \rangle, \end{aligned} \quad (8)$$

where $\dot{\gamma}_{z,r}(x) = \frac{d}{dr} \gamma_{z,r}(x)$ is the path derivative with respect to r . This inspires us to consider

$$W_{r,z,x} := \langle \dot{\gamma}_{z,r}(x), \nabla f(x_+) - \nabla f(\gamma_{z,r}(x)) \rangle, \quad (9)$$

so that

$$\nu(x) \propto q(x) \exp(\mathbb{E}_{r,z} W_{r,z,x}),$$

where the expectation $\mathbb{E}_{r,z}[\cdot]$ is taken over $r \sim \text{Unif}([0, 1])$ and $z \sim P$. Again, we will truncate $W_{r,z,x}$ to ensure that the estimator lies in $[-B, B]$.

For better dimension dependence, we consider the following path function ($\widehat{x} = x_0 - \eta \nabla f(x_+)$):

$$\begin{aligned} \gamma_{z,r}(x) &= a_r x + (1 - a_r) \widehat{x} + b_r z, \\ a_r &= \sin(\pi r/2), \quad b_r = \cos(\pi r/2), \end{aligned} \quad (10)$$

so that $\dot{\gamma}_{z,r}(x) = a'_r(x - \widehat{x}) + b'_r z$.

Assumption 3.2. There exists $s \in [0, 1]$ and $\beta_s \geq 0$ such that $\|\nabla f(x) - \nabla f(y)\| \leq \beta_s \|x - y\|^s$ for all $x, y \in \mathbb{R}^d$.

Theorem 3.3. Suppose that [Assumption 3.2](#) holds, $B = \Theta(1)$, and $\|x_0 - \eta \nabla f(x_+) - x_+\| \leq (d\eta)^{1/2}$.

Consider instantiating [Algorithm 1](#) with the choices $q = N(x_0 - \eta \nabla f(x_+), \eta I)$, and $\mathcal{W}_x$ the law of $\text{Clip}_B(W_{r,z,x})$, where $W_{r,z,x}$ is defined in (9)-(10) and $z \sim N(0, \eta I)$, $r \sim \text{Unif}([0, 1])$. Then, the law $\widehat{\nu}$ of [Algorithm 1](#) satisfies $D_{\chi^2}(\nu \parallel \widehat{\nu}) \leq \delta^2$, provided that

$$\eta^{-1} \gg \left(\beta_s^2 d^s \log(1/\delta) + \frac{s \beta_s^2}{d^{1-s}} \log^2(1/\delta) \right)^{1/(1+s)}.$$

Note that $x_0 - \eta \nabla f(x_+) - x_+ = 0$ when $x_+ = \text{prox}_{\eta f}(x_0)$, that is, the requirement of [Theorem 3.3](#) is that we can approximately take a proximal step on f from x_0 .

[Theorem 3.3](#) interpolates between the Lipschitz case $s = 0$, which requires $\eta^{-1} \gg \beta_0^2 \log(1/\delta)$, and the smooth case $s = 1$, which requires $\eta^{-1} \gg \beta_1 d^{1/2} \log(1/\delta)$. This recovers the result of [Fan et al. \(2023\)](#), except that we only use first-order queries; we spell out the implications for log-concave sampling in [Section 5](#).

4. Diffusion Sampling

A key insight is that sampling from the backward transition kernel (3), given access to score estimate s_t , is a special case of the setup of [Section 3](#), except that we only have approximately correct first-order evaluations. This leads to the template [Algorithm 2](#) for high-accuracy diffusion sampling. In the following subsections, we show how to choose q_t and $\mathcal{W}_x^t$ for $t \in [T]$.

Algorithm 2 Backward diffusion sampling

Input: Score estimates $\{s_t\}_{t \in [T]}$, initial distribution $\widehat{p}_T$, parameters $(\alpha_t, \eta_t)_{t \in [T]}$
 Sample $X_T \sim \widehat{p}_T$.
for $t = T - 1, \dots, 1$ **do**
 Sample $X_t \leftarrow \text{FORS}(B, q_t, (\mathcal{W}_x^t)_{x \in \mathbb{R}^d})$.
end for
Output: X_1

4.1. A Simple Method

We first propose a simple method that uses as a proposal the Gaussian portion of (3). We state all our guarantees for sampling from the early stopped distribution p_1 , the law of $X_1 \sim \mathcal{N}(\alpha_0 X_0, \sigma_1^2)$ with $X_0 \sim p_{\text{data}}$; see the discussion after the theorem.

Theorem 4.1. *Suppose that $\delta \in (0, \frac{1}{2}]$, $B = \Theta(1)$, and*

$$\frac{\sigma_t^2}{\eta_t} \gg d \log(1/\delta) + \log^2(1/\delta). \quad (11)$$

Consider instantiating the subroutine FORS in Algorithm 2 with the following choices: $q_t = \mathcal{N}(\bar{x}_t, \eta_t I)$, where $\bar{x}_t := \alpha_t^{-1} X_{t+1}$, and $\mathcal{W}_x$ is the law of

$$\widehat{W}_{r,x} := \text{Clip}_B(\langle x - \bar{x}_t, \mathbf{s}_t(r x + (1-r)\bar{x}_t) \rangle),$$

for $r \sim \text{Unif}([0, 1])$.

Let $\hat{p}_1$ be the law of X_1 generated by Algorithm 2. Then,

$$D_{\text{KL}}(p_1 \parallel \hat{p}_1) \lesssim D_{\text{KL}}(p_T \parallel \hat{p}_T) + T\delta + \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2.$$

Next, we describe the implications of Theorem 4.1 for diffusion model sampling. We should always choose $B = \Theta(1)$ and we denote $G := C(d + \log(1/\delta)) \log(d/\delta)$ for a sufficiently large constant C from Eq. (11). Then, in the variance-preserving setting:

- Recall that $\alpha_t^2 = \frac{1-\sigma_{t+1}^2}{1-\sigma_t^2}$ and $\sigma_{t+1}^2 = \frac{1-\sigma_{t+1}^2}{1-\sigma_t^2} (\sigma_t^2 + \eta_t)$.
- The condition $\eta_t \leq \frac{\sigma_t^2}{G}$ reduces to $\frac{\sigma_{t+1}^2}{1-\sigma_{t+1}^2} \leq \frac{\sigma_t^2}{1-\sigma_t^2} \cdot (1 + \frac{1}{G})$.
- This implies that as long as $T \geq O(G \log(1/(\delta \sigma_0^2)))$, we can guarantee that $1 - \sigma_T^2 \leq \bar{\delta}$.
- From known results on the convergence of the forward process, $1 - \sigma_T^2 \leq \bar{\delta}$ and $\hat{p}_T = \mathcal{N}(0, \sigma_T^2 I)$ imply that $D_{\text{KL}}(p_T \parallel \hat{p}_T) \lesssim \bar{\delta} \mathbb{E} \|X_0\|^2$.

By the above calculation, we can show the following corollary. We denote $M_2^2 := \mathbb{E}_{X_0 \sim p_{\text{data}}} \|X_0\|^2$ to be the second moment of p_{data} .

Corollary 4.2. *In the variance-preserving setting, for any $\delta \in (0, \frac{1}{2}]$, $\sigma_0 > 0$, there exists a schedule $(\sigma_t)_{t \in [T]}$ such that*

$$T \leq O((d + \log(1/\delta)) \log^2(\kappa/\delta)),$$

where $\kappa := (d + M_2^2)/\sigma_0^2$, and Algorithm 2 can be instantiated with $\hat{p}_T = \mathcal{N}(0, \sigma_T^2 I)$, so that

$$D_{\text{KL}}(p_1 \parallel \hat{p}_1) \lesssim \delta^2 + \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2.$$

Further, the number of queries made by Algorithm 2 is $O(T)$ with probability at least $1 - \delta$.

This governs the complexity of sampling from the early stopped distribution p_1 . To convert this into a guarantee for sampling from p_{data} itself, we can note that $W_2^2(p_{\text{data}}, p_1) \leq (1 - \alpha_0)^2 \mathbb{E} \|X_0\|^2 + \sigma_0^2 d$. This can be made at most δ^2 by choosing $\sigma_0^2 \asymp \delta^2/(d + M_2^2)$.

In terms of the bounded Lipschitz metric, this implies that $D_{\text{BL}}(p_{\text{data}}, \hat{p}_1)^2 \lesssim \delta^2 + \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2$ with a total complexity of

$$\max\{d, \log(1/\delta)\} \log^2\left(\frac{d + M_2^2}{\delta^2}\right).$$

Note that this guarantee imposes no assumptions on p_{data} beyond a second moment bound. This improves upon prior works Benton et al. (2024); Conforti et al. (2025) which achieved $\tilde{O}(d/\delta^2)$ under these assumptions, and Jain & Zhang (2025); Li & Yan (2025) which recently improved the complexity to $\tilde{O}(d/\delta)$.

Additionally, we can derive a TV guarantee assuming that the data-distribution p_{data} is log-smooth with parameter L . In this case, we can bound $D_{\chi^2}(p_{\text{data}} \parallel p_1) \leq e^{dL\sigma_0^2} - 1$ (Lemma B.14). Hence, we can choose $\sigma_0 \asymp \delta/(\sqrt{dL})$ to ensure that $D_{\text{KL}}(p_{\text{data}} \parallel \hat{p}_1) \lesssim \delta^2 + \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2$ with a total complexity of

$$\max\{d, \log(1/\delta)\} \log^2\left(\frac{d + M_2^2 + L}{\delta^2}\right).$$

4.2. A DDPM-Like Method

Next, we present a method which we call a ‘‘DDPM-like’’ method because the proposal q takes the form $\mathcal{N}(\alpha_t^{-1} X_{t+1} + \eta_t \mathbf{s}_t(\alpha_t^{-1} X_{t+1}), \eta_t I)$, which is similar but slightly different from a step of DDPM (5).

We show that our method, similar to the one from Section 3, achieves $\sqrt{d}$ -complexity under the *non-uniform Lipschitz condition*, following Jiao & Li (2024); Jiao et al. (2025).

Assumption 4.3 (Non-uniform Lipschitz condition). For any $\delta > 0$, there is a parameter $L_\delta \geq 1$ such that for any $t \in [T]$,

$$\mathbb{P}_{X_t \sim p_t} \left(\|\nabla \mathbf{s}_t^*(X_t)\|_{\text{op}} > \frac{L_\delta}{\sigma_t^2} \right) \leq \frac{\delta^2}{d^8}. \quad (12)$$

We note that by Proposition E.8, Assumption 4.3 holds unconditionally with $L_\delta = O(d + \log(1/\delta))$. In other words, the score function $\mathbf{s}_t^*$ is smooth ‘‘with high probability’’. Further, when p_{data} is Gaussian, Assumption 4.3 holds with $L_\delta \equiv 1$ for any $\delta \geq 0$. As shown in Jiao et al. (2025), when $p_{\text{data}} = \sum_{h=1}^H p_h \mathcal{N}(\mu_h, \sigma_h^2)$ is a mixture of H Gaussian

distributions, it holds that $L_\delta \leq O(\log(dH/\delta))$ (see also [Proposition E.7](#)).

Theorem 4.4. Suppose that [Assumption 4.3](#) holds, $\delta \in (0, \frac{1}{2}]$, $B = \Theta(1)$, and

$$\frac{\sigma_t^2}{\eta_t} \gg \sqrt{dL_\delta \log(d/\delta)} + L_\delta \log(d/\delta). \quad (13)$$

Consider instantiating the subroutine *FORS* in [Algorithm 2](#) with the following choices: $q_t = \mathcal{N}(\bar{x}_t + \eta_t \mathbf{s}_t(\bar{x}_t), \eta_t I)$, where $\bar{x}_t := \alpha_t^{-1} X_{t+1}$, and $\mathcal{W}_x^t$ is the law of

$$\widehat{W}_{r,z,x} := \text{Clip}_B(\langle \dot{\gamma}_{z,r,\bar{x}_t}(x), \mathbf{s}_t(\gamma_{z,r,\bar{x}_t}(x)) - \mathbf{s}_t(\bar{x}_t) \rangle),$$

where $z \sim \mathcal{N}(0, \eta_t I)$, $r \sim \text{Unif}([0, 1])$, and $\gamma_{z,r,\bar{x}_t}$ denotes the path function

$$\gamma_{z,r,\bar{x}_t}(x) := a_r x + (1 - a_r) \bar{x}_t + b_r z, \quad (14)$$

$$a_r := \sin(\pi r/2), \quad b_r := \cos(\pi r/2). \quad (15)$$

Let $\hat{p}_1$ be the law of X_1 generated by [Algorithm 2](#). Then

$$\begin{aligned} D_{\text{TV}}(p_1, \hat{p}_1)^2 &\leq 2D_H^2(p_1, \hat{p}_1) \\ &\lesssim D_{\text{KL}}(p_T \| \hat{p}_T) + T\delta + \sqrt{d/L_\delta} \cdot \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2. \end{aligned}$$

Following the reasoning of the preceding subsection, [Theorem 4.4](#) implies a complexity of

$$\boxed{\max\{\sqrt{dL_\delta \log(d/\delta)}, L_\delta \log(d/\delta)\} \log \frac{d + M_2^2}{\delta^2}}$$

to reach $D_{\text{BL}}(\hat{p}_1, p_{\text{data}})^2 \lesssim \delta^2 + \sqrt{d/L_\delta} \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2$. Note that the dependence on the score error is worse; we do not know if this is fundamental.

4.3. Adaptation to Low-Dimensional Structure

While [Theorem 4.4](#) improves the dependence on the ambient dimension to $\sqrt{d}$, ideally, we would like to choose η according to the *intrinsic dimension* of the data distribution p_{data} ([Li & Yan, 2024](#)) (cf. [Definition 4.6](#)).

The key observation is that for $p = p_{\text{data}} * \mathcal{N}(0, \sigma^2 I)$, we can write, for $X_0 \sim p_{\text{data}}$, $X \sim \mathcal{N}(X_0, \sigma^2 I)$,

$$\nabla^2 \log p(x) = -\frac{I}{\sigma^2} + \frac{\text{Cov}(X_0 | X = x)}{\sigma^4}, \quad (16)$$

and hence, when p_{data} has low-dimensional structure, we may expect the shifted Hessian $\nabla^2 \log p + \sigma^{-2} I$ to be well-behaved. This motivates adapting [Theorem 4.4](#) by replacing the score function with the shifted score function $\tilde{\mathbf{s}}(x) = \mathbf{s}(x) + \sigma^{-2} x$.

Definition 4.5 (Covering number). For any distribution $\bar{p}$ and $r \geq 0$, let $N(\bar{p}; r)$ be the r -covering number of $\text{supp}(\bar{p})$ under Euclidean norm $\|\cdot\|$.

Definition 4.6 (Intrinsic dimension). Suppose that the data distribution p_{data} has support $\bar{\mathcal{X}} = \text{supp}(p_{\text{data}})$. Let $N_{\text{data}}(r) = N(\bar{\mathcal{X}}; r, \|\cdot\|_2)$ be the covering number of $\bar{\mathcal{X}}$ under Euclidean norm, and we define the *intrinsic dimension* of p_{data} as $d_\star = 1 + \log N_{\text{data}}(\sigma_0/\sqrt{d})$.

Theorem 4.7. Suppose that $\delta \in (0, \frac{1}{2}]$, $B = \Theta(1)$, and

$$\frac{\sigma_t^2}{\eta_t} \gg d_\star \log(d/\delta) + \log^2(d/\delta). \quad (17)$$

Consider instantiating the subroutine *FORS* in [Algorithm 2](#) with the following choices: $q_t = \mathcal{N}(\bar{x}_t + \bar{\eta}_t \mathbf{s}_t(\bar{x}_t), \bar{\eta}_t I)$, where $\bar{x}_t := \alpha_t^{-1} X_{t+1}$, $\lambda_t = \frac{1}{\sigma_t^2}$, $1/\bar{\eta}_t = 1/\eta_t + \lambda_t$, and $\mathcal{W}_x^t$ is the law of

$$\begin{aligned} \widehat{W}_{r,z,x} &:= \text{Clip}_B(\langle \dot{\gamma}_{z,r,\bar{x}_t}(x), \mathbf{s}_t(\gamma_{z,r,\bar{x}_t}(x)) - \mathbf{s}_t(\bar{x}_t) \\ &\quad + \lambda_t (\gamma_{z,r,\bar{x}_t}(x) - \bar{x}_t) \rangle), \end{aligned}$$

where $z \sim \mathcal{N}(0, \eta_t I)$, $r \sim \text{Unif}([0, 1])$, and $\gamma_{z,r,\bar{x}_t}$ is the path function given in (14)–(15).

Let $\hat{p}_1$ be the law of X_1 generated by [Algorithm 2](#). Then

$$D_{\text{KL}}(p_1 \| \hat{p}_1) \lesssim D_{\text{KL}}(p_T \| \hat{p}_T) + T\delta + \frac{d}{d_\star} \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2,$$

where for each $t \in [T]$, we define

$$\bar{\varepsilon}_{t,\text{score}}^2 := \max_{\eta \in [0, 2\eta_t]} \mathbb{E}_{X \sim p_t * \mathcal{N}(0, \eta I)} \|\mathbf{s}_t(X_t) - \mathbf{s}_t^\star(X_t)\|^2.$$

This implies a complexity of

$$\boxed{\max\{d_\star, \log(d/\delta)\} \log^2\left(\frac{d + M_2^2}{\delta^2}\right)}$$

to reach $D_{\text{BL}}(p_{\text{data}}, \hat{p}_1)^2 \lesssim \delta^2 + (d/d_\star) \sum_{t=1}^{T-1} \eta_t \varepsilon_{t,\text{score}}^2$. Again, the dependence on the score error is worse.

5. Application to Log-Concave Sampling

In this section, we briefly describe the implications of our results ([Theorem 3.3](#)) for the problem of sampling from a density $\mu \propto e^{-f}$ under log-concavity and, more generally, isoperimetric conditions. The key connection lies in the *proximal sampler* ([Lee et al., 2021](#); [Chen et al., 2022](#)), which is an instance of the Gibbs sampler in which the target distribution is the augmented density

$$\tilde{\mu}(x, y) \propto \exp\left(-f(x) - \frac{1}{2\eta} \|y - x\|^2\right).$$

Algorithm 3 Proximal sampler

Input: Gradient ∇f , step size $\eta > 0$
for $n = 1, 2, 3, \dots, N$ **do**
 Sample $Y_n \sim \mathcal{N}(X_n, \eta I)$.
 Sample $X_{n+1} \sim \text{RGO}_{f, \eta, Y_n}$.
end for
 Output X_N .

See [Algorithm 3](#).

[Algorithm 3](#) hinges on being able to implement the *restricted Gaussian oracle* (RGO):

$$\text{RGO}_{f, \eta, y}(x) \propto_x \exp\left(-f(x) - \frac{1}{2\eta} \|y - x\|^2\right).$$

Convergence of [Algorithm 3](#) was shown in [Lee et al. \(2021\)](#) under log-concavity of μ , and in [Chen et al. \(2022\)](#) under weaker isoperimetric assumptions. On the other hand, note that the RGO is exactly a Gaussian tilt distribution, which we considered in [Section 3](#). Hence, implementing the RGO step using FORS leads to novel high-accuracy sampling results for log-concave (and isoperimetric) distributions, *without* assuming access to zeroth-order queries for f . Almost all previous approaches (e.g., rejection sampling, Metropolis–Hastings) require zeroth-order queries, whereas algorithms using only first-order queries typically arise as discretizations of diffusion processes and the resulting discretization error degrades their accuracy guarantees. A notable **exception** is the result of [Lu & Wang \(2022\)](#) on the zigzag sampler, an example of a PDMP, which achieved a complexity of $\tilde{O}(\kappa^2 d^{3/2} \log^{5/2}(1/\delta))$ evaluations of *partial* derivatives of f , under strong-log-concavity and given a warm start, where κ is the condition number.

We only summarize some representative results here, and defer a full presentation to [Appendix F](#). Let $\mu_0, \hat{\mu}$ denote the initialization and output of [Algorithm 3](#), and in all cases we track the number of queries to a first-order and proximal oracle for f in expectation.

Suppose that f is smooth ($s = 1$ in [Assumption 3.2](#)).

- Under a log-Sobolev inequality ([Definition B.2](#)), we obtain $D_{\chi^2}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ in $\tilde{O}(\kappa(d^{1/2} \log^{3/2}(\mathcal{R}/\varepsilon^2) + \log^2(\mathcal{R}/\varepsilon^2)))$ queries, where $\mathcal{R} := \log(1 + D_{\chi^2}(\mu_0 \parallel \mu))$ and $\kappa := C_{\text{LSI}} \beta_1$.
- Under a Poincaré inequality ([Definition B.1](#)), we obtain $D_{\chi^2}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ in $\tilde{O}(\kappa(d^{1/2} \log^{1/2}(1/\varepsilon) + \log(1/\varepsilon)) \log(\chi^2/\varepsilon^2))$ queries, where $\chi^2 := D_{\chi^2}(\mu_0 \parallel \mu)$ and $\kappa := C_{\text{PI}} \beta_1$.
- Under log-concavity, we obtain $D_{\text{KL}}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ in $\tilde{O}(\beta_1 d^{1/2} W_2^2(\mu_0, \mu)/\varepsilon^2)$ queries. In addition, recent progress toward the KLS conjecture ([Klartag, 2023](#)) im-

plies $C_{\text{PI}}(\mu) \leq O(\log d) \cdot \|\mathbb{E}_\mu[XX^\top]\|_{\text{op}}$ under log-concavity, so the previous result also yields a high-accuracy sampling guarantee for any log-concave μ (albeit with dependence on $\log \chi^2$).

Suppose now that f is Lipschitz ($s = 0$ in [Assumption 3.2](#)).

- Under a Poincaré inequality, we obtain $D_{\chi^2}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ in $\tilde{O}(C_{\text{PI}} \beta_0^2 \log(1/\varepsilon) \log(\chi^2/\varepsilon^2))$ queries.
- Under log-concavity, we obtain $D_{\text{KL}}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ in $\tilde{O}(\beta_0^2 W_2^2(\mu_0, \mu)/\varepsilon^2)$ queries.

6. Conclusion

In this work, we have presented high-accuracy algorithms for diffusion sampling which operate under minimal data assumptions ([Theorem 4.1](#)), with improved dimension dependence under a Lipschitz score assumption ([Theorem 4.4](#)) and low-dimensional data ([Theorem 4.7](#)). With regards to the dependence on the target accuracy, our results improve exponentially over all prior works. Our framework also has implications for log-concave sampling ([Section 5](#)) using only first-order queries.

Although this is primarily a theoretical work, we believe it opens a path toward practical improvements. FORS can be viewed as a principled accept-reject correction mechanism that relies only on score evaluations, suggesting that one may design lightweight corrector steps—inspired by our rejection sampling procedure—to improve sample quality in large-step size regimes where standard DDPM begins to lose accuracy. However, there are several practical challenges. While our framework allows larger step sizes, the FORS subroutine requires multiple score evaluations per step—albeit only logarithmically many—which increases the per-step cost. Additionally, the step size conditions in our theory may be conservative in practice. Translating our algorithm into a computationally competitive practical method remains an important direction for future work.

Finally, we note that ODE-based samplers such as DDIM can perform well in practice with substantially fewer steps than SDE-based samplers, but without a theoretical high-accuracy guarantee. The focus of this work is on accelerating SDE-based samplers, and obtaining analogous high-accuracy guarantees for ODE-based methods is an important open problem.

Impact Statement

Our work provides new algorithms for diffusion sampling with potential speed-ups. This could lead to many potential impacts, none of which we feel the need to specifically highlight here.

Acknowledgments

We thank Sam Power for bringing to our attention useful references. We acknowledge support from ARO through award W911NF-21-1-0328, Simons Foundation, and the NSF through awards DMS-2031883 and PHY-2019786, the DARPA AIQ program, and AFOSR FA9550-25-1-0375. CD is supported by a Simons Investigator Award, a Simons Collaboration on Algorithmic Fairness, ONR MURI grant N00014-25-1-2116, and ONR grant N00014-25-1-2296.

References

- Altschuler, J. M. and Chewi, S. Faster high-accuracy log-concave sampling via algorithmic warm starts. *J. ACM*, 71(3), 6 2024.
- Azangulov, I., Deligiannidis, G., and Rousseau, J. Convergence of diffusion models under the manifold hypothesis in high-dimensions. *arXiv preprint arXiv:2409.18804*, 2024.
- Bakry, D., Gentil, I., and Ledoux, M. *Analysis and geometry of Markov diffusion operators*, volume 348 of *Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer, Cham, 2014.
- Benton, J., Bortoli, V. D., Doucet, A., and Deligiannidis, G. Nearly d -linear convergence bounds for diffusion models via stochastic localization. In *The Twelfth International Conference on Learning Representations*, 2024.
- Beskos, A. and Roberts, G. O. Exact simulation of diffusions. 2005.
- Beskos, A., Papaspiliopoulos, O., and Roberts, G. O. Retrospective exact simulation of diffusion sample paths with applications. *Bernoulli*, 12(6):1077–1098, 2006.
- Chen, H., Lee, H., and Lu, J. Improved analysis of score-based generative modeling: user-friendly bounds under minimal smoothness assumptions. In *International Conference on Machine Learning*, pp. 4735–4763. PMLR, 2023a.
- Chen, S., Chewi, S., Lee, H., Li, Y., Lu, J., and Salim, A. The probability flow ODE is provably fast. In Oh, A., Neumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S. (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 68552–68575. Curran Associates, Inc., 2023b.
- Chen, S., Chewi, S., Li, J., Li, Y., Salim, A., and Zhang, A. R. Sampling is as easy as learning the score: theory for diffusion models with minimal data assumptions. In *The Eleventh International Conference on Learning Representations*, 2023c.
- Chen, Y., Chewi, S., Salim, A., and Wibisono, A. Improved analysis for a proximal algorithm for sampling. In Loh, P.-L. and Raginsky, M. (eds.), *Proceedings of Thirty Fifth Conference on Learning Theory*, volume 178 of *Proceedings of Machine Learning Research*, pp. 2984–3014. PMLR, 7 2022.
- Chewi, S. *Log-concave sampling*. Forthcoming, 2026. Available online at <https://chewisinho.github.io/>.
- Chewi, S., Erdogdu, M. A., Li, M. B., Shen, R., and Zhang, M. S. Analysis of Langevin Monte Carlo from Poincaré to log-Sobolev. *Found. Comput. Math.*, 2024.
- Conforti, G., Durmus, A., and Gentiloni Silveri, M. KL convergence guarantees for score diffusion models under minimal data assumptions. *SIAM J. Math. Data Sci.*, 7(1):86–109, 2025.
- Duchi, J. C. Lecture notes on statistics and information theory. 2023.
- Fan, J., Yuan, B., and Chen, Y. Improved dimension dependence of a proximal algorithm for sampling. In Neu, G. and Rosasco, L. (eds.), *Proceedings of Thirty Sixth Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pp. 1473–1521. PMLR, 7 2023.
- Foster, D. J., Han, Y., Qian, J., and Rakhlin, A. Online estimation via offline estimation: an information-theoretic framework. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 42840–42898. Curran Associates, Inc., 2024.
- Gao, X. and Zhu, L. Convergence analysis for general probability flow ODEs of diffusion models in Wasserstein distances. In Li, Y., Mandt, S., Agrawal, S., and Khan, E. (eds.), *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pp. 1009–1017. PMLR, 5 2025.
- Gao, X., Nguyen, H. M., and Zhu, L. Wasserstein convergence guarantees for a general class of score-based generative models. *Journal of Machine Learning Research*, 26(43):1–54, 2025.
- Gatmiry, K., Chen, S., and Salim, A. High-accuracy and dimension-free sampling with diffusions. *arXiv preprint arXiv:2601.10708*, 2026.
- Huang, D. Z., Huang, J., and Lin, Z. Convergence analysis of probability flow ODE for score-based generative models. *IEEE Transactions on Information Theory*, 2025a.

- Huang, D. Z., Huang, J., and Lin, Z. Fast convergence for high-order ODE solvers in diffusion probabilistic models. *arXiv preprint arXiv:2506.13061*, 2025b.
- Huang, X., Zou, D., Dong, H., Zhang, Z., Ma, Y., and Zhang, T. Reverse transition kernel: a flexible framework to accelerate diffusion inference. *Advances in Neural Information Processing Systems*, 37:95515–95578, 2024a.
- Huang, X., Lin, Y., Kuang, N. L., Dong, H., Zou, D., Ma, Y., and Zhang, T. Almost linear convergence under minimal score assumptions: quantized transition diffusion, 2025c.
- Huang, Z., Wei, Y., and Chen, Y. Denoising diffusion probabilistic models are optimally adaptive to unknown low dimensionality. *arXiv preprint arXiv:2410.18784*, 2024b.
- Jain, N. and Zhang, T. A sharp KL-convergence analysis for diffusion models under minimal assumptions. *arXiv preprint 2508.16306*, 2025.
- Jain, N. and Zhang, T. A sharp KL convergence analysis for diffusion models under minimal assumptions. In *The Fourteenth International Conference on Learning Representations*, 2026.
- Jayram, T. S. Hellinger strikes back: a note on the multi-party information complexity of AND. In *Approximation, randomization, and combinatorial optimization*, volume 5687 of *Lecture Notes in Comput. Sci.*, pp. 562–573. Springer, Berlin, 2009.
- Jiao, Y. and Li, G. Instance-dependent convergence theory for diffusion models. *arXiv preprint arXiv:2410.13738*, 2024.
- Jiao, Y., Zhou, Y., and Li, G. Optimal convergence analysis of DDPM for general distributions. *arXiv preprint arXiv:2510.27562*, 2025.
- Keane, M. and O’Brien, G. L. A Bernoulli factory. *ACM Transactions on Modeling and Computer Simulation (TOMACS)*, 4(2):213–219, 1994.
- Klartag, B. Logarithmic bounds for isoperimetry and slices of convex sets. *arXiv preprint arXiv:2303.14938*, 2023.
- Lee, H., Lu, J., and Tan, Y. Convergence of score-based generative modeling for general data distributions. In Agrawal, S. and Orabona, F. (eds.), *Proceedings of the 34th International Conference on Algorithmic Learning Theory*, volume 201 of *Proceedings of Machine Learning Research*, pp. 946–985. PMLR, 2 2023.
- Lee, Y. T., Shen, R., and Tian, K. Structured logconcave sampling with a restricted Gaussian oracle. In Belkin, M. and Kpotufe, S. (eds.), *Proceedings of Thirty Fourth Conference on Learning Theory*, volume 134 of *Proceedings of Machine Learning Research*, pp. 2993–3050. PMLR, 8 2021.
- Li, G. and Cai, C. Provable acceleration for diffusion models under minimal assumptions. *arXiv preprint arXiv:2410.23285*, 2024.
- Li, G. and Yan, Y. Adapting to unknown low-dimensional structures in score-based diffusion models. *Advances in Neural Information Processing Systems*, 37:126297–126331, 2024.
- Li, G. and Yan, Y. $O(d/T)$ convergence theory for diffusion probabilistic models under minimal assumptions. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Li, G., Wei, Y., Chen, Y., and Chi, Y. Towards faster non-asymptotic convergence for diffusion-based generative models. *arXiv preprint arXiv:2306.09251*, 2023.
- Li, G., Huang, Y., Efimov, T., Wei, Y., Chi, Y., and Chen, Y. Accelerating convergence of score-based diffusion models, provably. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 27942–27954. PMLR, 7 2024a.
- Li, G., Wei, Y., Chi, Y., and Chen, Y. A sharp convergence theory for the probability flow ODEs of diffusion models. *arXiv preprint arXiv:2408.02320*, 2024b.
- Li, G., Cai, C., and Wei, Y. Dimension-free convergence of diffusion models for approximate Gaussian mixtures. *arXiv preprint arXiv:2504.05300*, 2025a.
- Li, G., Zhou, Y., Wei, Y., and Chen, Y. Faster diffusion models via higher-order approximation. *arXiv preprint arXiv:2506.24042*, 2025b.
- Liang, J., Huang, Z., and Chen, Y. Low-dimensional adaptation of diffusion models: convergence in total variation. *arXiv preprint arXiv:2501.12982*, 2025.
- Lu, J. and Wang, L. Complexity of zigzag sampling algorithm for strongly log-concave distributions. *Stat. Comput.*, 32(3):Paper No. 48, 12, 2022.
- Nacu, Ș. and Peres, Y. Fast simulation of new coins from old. 2005.
- Papaspiliopoulos, O. Monte Carlo probabilistic inference for diffusion processes: a methodological framework. In *Bayesian time series models*, pp. 82–103. Cambridge Univ. Press, Cambridge, 2011.

- Potapchik, P., Azangulov, I., and Deligiannidis, G. Linear convergence of diffusion models under the manifold hypothesis. *arXiv preprint arXiv:2410.09046*, 2024.
- Tang, J. and Yan, Y. Adaptivity and convergence of probability flow ODEs in diffusion generative models. *arXiv preprint arXiv:2501.18863*, 2025.
- Wagner, W. Monte Carlo evaluation of functionals of solutions of stochastic differential equations. Variance reduction and numerical examples. *Stochastic Anal. Appl.*, 6 (4):447–468, 1988.
- Wainwright, M. J. Score-based sampling without diffusions: guidance from a simple and modular scheme. *arXiv preprint arXiv:2512.24152*, 2025.
- Xun, Z. and Price, E. Query lower bounds for diffusion sampling. *arXiv preprint arXiv:2604.10857*, 2026.
- Zhang, M. S., Huan, S., Huang, J., Boffi, N. M., Chen, S., and Chewi, S. Sublinear iterations can suffice even for DDPMs. *arXiv preprint arXiv:2511.04844*, 2025.

Contents

1	Introduction	1
1.1	Our Contribution	2
1.2	Related Work	2
1.3	Notation	3
2	Background on Diffusion Models	3
3	Key Subroutine: Gaussian Tilts	4
3.1	First-Order Rejection Sampling (FORS)	4
3.2	Sampling from Gaussian Tilts with an Exact Oracle	4
4	Diffusion Sampling	5
4.1	A Simple Method	6
4.2	A DDPM-Like Method	6
4.3	Adaptation to Low-Dimensional Structure	7
5	Application to Log-Concave Sampling	7
6	Conclusion	8
A	Discussion	12
A.1	Concurrent Work	12
B	Additional Notation and Technical Tools	13
B.1	Functional Inequalities	13
B.2	Lemmas	13
C	Proof of Theorem 3.1	18
D	Proofs for Section 3	18
E	Proofs for Section 4	20
E.1	Proof of Theorem 4.1	20
E.2	Analysis of a DDPM-Like Method	22
E.3	Proof of Theorem 4.4	23
E.4	Proof of Theorem 4.7	25
E.5	Technical Lemmas	27
E.6	Proof of Proposition E.3	29
E.7	Proof of Proposition E.4	31
F	Log-Concave Sampling	34

A. Discussion

A.1. Concurrent Work

In independent and concurrent work, [Gatmiry et al. \(2026\)](#) study an accelerated ODE flow and remarkably obtain a *high-accuracy* guarantee under a set of structural assumptions. This is highly non-trivial since, as discussed in the introduction, prior works based on higher-order methods incur implicit exponential dependencies on problem parameters and the order of the method. Their results rely on the following conditions:

(A1) The data distribution satisfies $p_{\text{data}} = p_{\star} * \mathcal{N}(0, \sigma^2 I)$, where $p_{\star}$ is supported on a ball of radius R .

(A2) *Sub-exponential score error*: for some $\varepsilon_{\text{score}} = \tilde{O}(\delta)$,

$$\mathbb{P}_{X \sim p_t}(\|\mathbf{s}_t(X) - \mathbf{s}_t^*(X)\| \geq u) \leq \exp(-u/\varepsilon_{\text{score}}).$$

(A3) The score estimates $(s_t)_{t \in [T]}$ are Lipschitz.

Under these assumptions, [Gatmiry et al. \(2026\)](#) propose a method with query complexity $\tilde{O}((R/\sigma)^2 \log^2(1/\delta))$.

In comparing with our results, we make the following remarks:

- The query complexity, which scales with R/σ , is incomparable to ours. On one hand, when R/σ is constant, then their guarantee is *dimension-free*. On the other, R/σ could be large in many settings.

For example, as noted by [Gatmiry et al. \(2026\)](#), (A1) indeed holds in the *early stopping* regime, i.e., their data distribution actually corresponds to the early stopped distribution p_1 . However, for this case, achieving distributional closeness to the real data distribution p_* typically requires $\sigma \ll \delta/\sqrt{d}$, and then we should expect $\text{poly}(1/\delta)$ dependence, instead of a high-accuracy $\text{polylog}(1/\delta)$ dependence. Further, the bound R may also scale with $\sqrt{d}$ in some settings.

- Assumption (A2) is substantially stronger than our average error condition ([Definition 2.1](#)). It is unclear whether standard statistical learning procedures can achieve convergence guarantees expressed in terms of sub-exponential tail bounds.

Finally, while their analysis is specific to accelerated ODE-based diffusion models, our approach applies more broadly to general first-order sampling methods, including log-concave sampling ([Section 5](#)).

B. Additional Notation and Technical Tools

For any function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $Z_f := \int_{\mathbb{R}^d} e^{-f(x)} dx < +\infty$, we define μ_f to be the distribution over $\mathbb{R}^d$ with density $\mu_f(x) = \frac{1}{Z_f} e^{-f(x)}$.

For $\nu \ll \mu$, let $\mathcal{R}_\lambda(\nu \parallel \mu) := \frac{1}{\lambda-1} \log \mathbb{E}_\mu[(\frac{d\nu}{d\mu})^\lambda]$ denote the Rényi divergence of order $\lambda > 1$. In addition, we define

$$D_\lambda(p \parallel q) := \mathbb{E}_p\left(\frac{dp}{dq}\right)^\lambda - 1, \quad \bar{D}_\lambda(p \parallel q) = \max\{D_\lambda(p \parallel q), D_\lambda(q \parallel p)\}. \quad (18)$$

B.1. Functional Inequalities

Definition B.1 (PI). A probability distribution μ on $\mathbb{R}^d$ satisfies a Poincaré inequality (PI) with constant C if for all smooth and compactly supported functions $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\text{Var}_\mu(\phi) \leq C \mathbb{E}_\mu \|\nabla \phi\|^2.$$

We let $C_{\text{PI}}(\mu)$ to be the smallest constant C such that μ satisfies PI with constant C .

Definition B.2 (LSI). A probability distribution μ on $\mathbb{R}^d$ satisfies a log-Sobolev inequality (LSI) with constant C if for all smooth and compactly supported functions $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\mathbb{E}_\mu \left[\phi^2 \log \frac{\phi^2}{\mathbb{E}_\mu[\phi^2]} \right] \leq 2C \mathbb{E}_\mu \|\nabla \phi\|^2.$$

We let $C_{\text{LSI}}(\mu)$ to be the smallest constant C such that μ satisfies LSI with constant C .

The following facts are standard, see [Bakry et al. \(2014\)](#) or [Chewi \(2026, Chapter 2\)](#).

Lemma B.3 (Bakry–Émery). If μ is α -strongly log-concave, then $C_{\text{LSI}}(\mu) \leq \frac{1}{\alpha}$.

Lemma B.4 (Holley–Stroock perturbation principle). Let μ and ν be probability measures such that $\frac{1}{b} \leq \frac{d\nu}{d\mu} \leq b$ for some parameter $b > 0$. Then $C_{\text{LSI}}(\nu) \leq b^2 C_{\text{LSI}}(\mu)$.

B.2. Lemmas

Lemma B.5.

$$\mathbb{E}_{W \sim \mathcal{N}(0, \eta I)} \exp(|\langle W, v \rangle|) \leq 2 \exp\left(\frac{\eta \|v\|^2}{2}\right).$$

For $\lambda \leq \frac{1}{2\eta}$,

$$\mathbb{E}_{W \sim \mathcal{N}(0, \eta I)} \exp\left(\frac{1}{2} \lambda \|W + v\|^2\right) \leq \exp(d\lambda\eta + \lambda \|v\|^2).$$

Further, for $t \geq 0$,

$$\mathbb{P}_{x \sim \mathcal{N}(v, \Sigma)}(|x^\top Ax - v^\top Av - \text{tr}(\Sigma A)| \geq t) \leq 2 \exp\left(-\frac{1}{16\|\Sigma\|_{\text{op}}} \min\left\{\frac{t^2}{\|\Sigma\|_{\text{op}}\|A\|_{\text{F}}^2 + \|Av\|^2}, \frac{t}{\|A\|_{\text{op}}}\right\}\right).$$

Proof. The first two statements are standard. For the third, for any $A \preceq \frac{1}{2}\Sigma^{-1}$, we can calculate

$$\mathbb{E}_{x \sim \mathcal{N}(v, \Sigma)} \exp\left(\frac{1}{2} x^\top Ax - \frac{1}{2} v^\top Av\right) = \mathbb{E}_{W \sim \mathcal{N}(0, \Sigma)} \exp\left(v^\top AW + \frac{1}{2} W^\top AW\right) = \sqrt{\frac{\det(\tilde{\Sigma})}{\det(\Sigma)}} \exp\left(\frac{1}{2} \|Av\|_{\tilde{\Sigma}}^2\right),$$

where $\tilde{\Sigma}^{-1} = \Sigma^{-1} - A$. Noting that $\Sigma^{1/2} \tilde{\Sigma}^{-1} \Sigma^{1/2} = I - \Sigma^{1/2} A \Sigma^{1/2} \succeq \frac{1}{2} I$, we can bound

$$\frac{\det(\tilde{\Sigma})}{\det(\Sigma)} = \frac{1}{\det(I - \Sigma^{1/2} A \Sigma^{1/2})} \leq \exp(\text{tr}(\Sigma^{1/2} A \Sigma^{1/2}) + \|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{F}}^2),$$

where we use $-\log(1 - \lambda) \leq \lambda + \lambda^2$ for $\lambda \leq \frac{1}{2}$. Therefore, we have shown

$$\mathbb{E}_{x \sim \mathcal{N}(v, \Sigma)} \exp\left(\frac{1}{2} (x^\top Ax - v^\top Av - \text{tr}(\Sigma A))\right) \leq \exp(\|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{F}}^2 + \|Av\|_{\Sigma}^2),$$

as long as $A \preceq \frac{1}{2}\Sigma^{-1}$. Then, by rescaling $A \leftarrow \lambda A$ and requiring $|\lambda| \leq \frac{1}{2\|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{op}}}$, we have

$$\mathbb{P}(|x^\top Ax - v^\top Av - \text{tr}(\Sigma A)| \geq t) \leq 2 \exp\left(\lambda^2 (\|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{F}}^2 + \|Av\|_{\Sigma}^2) - \frac{1}{2} \lambda t\right), \quad \forall t > 0.$$

Suitably choosing λ gives, for all $t \geq 0$,

$$\begin{aligned} \mathbb{P}(|x^\top Ax - v^\top Av - \text{tr}(\Sigma A)| \geq t) &\leq 2 \exp\left(-\min\left\{\frac{t^2}{16(\|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{F}}^2 + \|Av\|_{\Sigma}^2)}, \frac{t}{8\|\Sigma^{1/2} A \Sigma^{1/2}\|_{\text{op}}}\right\}\right) \\ &\leq 2 \exp\left(-\frac{1}{16\|\Sigma\|_{\text{op}}} \min\left\{\frac{t^2}{\|\Sigma\|_{\text{op}}\|A\|_{\text{F}}^2 + \|Av\|^2}, \frac{t}{\|A\|_{\text{op}}}\right\}\right). \end{aligned}$$

□

Lemma B.6. Suppose that $\eta > 0$ and $0 \leq \lambda \leq \frac{d^{1-s}}{4s\eta^s}$. Then, it holds that

$$\mathbb{E}_{W \sim \mathcal{N}(0, \eta I)} \exp(\lambda \|W\|^{2s}) \leq \exp(2(\eta d)^s \lambda).$$

Proof. We use the inequality $w^s \leq s \cdot \frac{w}{(\eta d)^{1-s}} + (1-s) \cdot (\eta d)^s$ for $w \geq 0$. Therefore, as long as $\frac{s\lambda}{(\eta d)^{1-s}} \leq \frac{1}{4\eta}$, it holds that

$$\mathbb{E} \exp(\lambda \|W\|^{2s}) \leq \mathbb{E} \exp\left(\frac{s\lambda}{(\eta d)^{1-s}} \|W\|^2 + (1-s)(\eta d)^s \lambda\right) \leq \exp(2s(\eta d)^s \lambda + (1-s)(\eta d)^s \lambda) \leq \exp(2(\eta d)^s \lambda),$$

where the second inequality follows from [Lemma B.5](#). □

Lemma B.7. For any $f : \mathcal{X} \rightarrow [0, 1]$, it holds that

$$\mathbb{E}_p[f] - M \mathbb{E}_q[f] \leq \inf_{\lambda > 1} M^{-(\lambda-1)} (1 + D_\lambda(p \| q)).$$

Proof. By definition,

$$\mathbb{E}_p[f] - M \mathbb{E}_q[f] = \mathbb{E}_q \left[\left(\frac{dp}{dq} - M \right) f \right] \leq \mathbb{E}_q \left(\frac{dp}{dq} - M \right)_+.$$

Note that for any random variable $Y \geq 0$, we have

$$\mathbb{E}(Y - M)_+ = \mathbb{E}[\mathbb{I}\{Y > M\} (Y - M)_+] \leq \mathbb{P}(Y > M)^{1-\frac{1}{\lambda}} \left(\mathbb{E}[(Y - M)_+^\lambda] \right)^{\frac{1}{\lambda}} \leq \frac{\mathbb{E}[Y^\lambda]}{M^{\lambda-1}}.$$

Combining these inequalities and taking infimum over $\lambda > 1$ completes the proof. $\square$

Lemma B.8.

$$D_{\text{KL}}(\nu \parallel \hat{\mu}) \leq 2D_{\text{KL}}(\nu \parallel \mu) + \log(1 + D_{\chi^2}(\mu \parallel \hat{\mu})).$$

Proof. We define $h = \log \frac{d\mu}{d\hat{\mu}}$. Then, we know that $\mathbb{E}_\mu[e^h] = 1 + D_{\chi^2}(\mu \parallel \hat{\mu})$, and hence

$$\begin{aligned} D_{\text{KL}}(\nu \parallel \hat{\mu}) - D_{\text{KL}}(\nu \parallel \mu) &= \mathbb{E}_\nu \left[\log \frac{d\nu}{d\hat{\mu}} - \log \frac{d\nu}{d\mu} \right] = \mathbb{E}_\nu[h] \\ &\leq D_{\text{KL}}(\nu \parallel \mu) + \log \mathbb{E}_\mu[e^h] = D_{\text{KL}}(\nu \parallel \mu) + \log(1 + D_{\chi^2}(\mu \parallel \hat{\mu})), \end{aligned}$$

where we use the Donsker–Varadhan variational inequality. $\square$

Lemma B.9. Suppose that $\mu \propto pe^h$ and $\hat{\mu} \propto qe^h$, such that $|h| \leq B$. Then it holds that

$$D_{\chi^2}(\mu \parallel \hat{\mu}) \leq e^{4B} D_{\chi^2}(p \parallel q).$$

Proof. By definition,

$$1 + D_{\chi^2}(\mu \parallel \hat{\mu}) = \mathbb{E}_{x \sim \mu} \left[\frac{\mu(x)}{\hat{\mu}(x)} \right] = \frac{\mathbb{E}_q[e^h]}{(\mathbb{E}_p[e^h])^2} \cdot \mathbb{E}_{x \sim q} \left[\frac{p(x)^2}{q(x)^2} \cdot e^{h(x)} \right].$$

Note that $\frac{p(x)^2}{q(x)^2} = \left(\frac{p(x)}{q(x)} - 1 \right)^2 + 2 \frac{p(x)}{q(x)} - 1$, and hence

$$\mathbb{E}_{x \sim q} \left[\frac{p(x)^2}{q(x)^2} \cdot e^{h(x)} \right] = \mathbb{E}_{x \sim q} \left[\left(\frac{p(x)}{q(x)} - 1 \right)^2 \cdot e^{h(x)} \right] + 2 \mathbb{E}_p[e^h] - \mathbb{E}_q[e^h].$$

Hence, we can rewrite

$$\begin{aligned} D_{\chi^2}(\mu \parallel \hat{\mu}) &= \frac{\mathbb{E}_q[e^h]}{(\mathbb{E}_p[e^h])^2} \cdot \mathbb{E}_{x \sim q} \left[\left(\frac{p(x)}{q(x)} - 1 \right)^2 \cdot e^{h(x)} \right] - \left(\frac{\mathbb{E}_q[e^h]}{\mathbb{E}_p[e^h]} - 1 \right)^2 \\ &\leq e^{4B} \mathbb{E}_{x \sim q} \left[\left(\frac{p(x)}{q(x)} - 1 \right)^2 \right] = e^{4B} D_{\chi^2}(p \parallel q). \end{aligned}$$

$\square$

Lemma B.10. Suppose that $P, Q \in \Delta(\mathcal{Z})$. Then, for $h : \mathcal{Z} \rightarrow [0, 1]$,

$$\mathbb{E}_P[h] \leq 3 \mathbb{E}_Q[h] + 4D_{\text{H}}^2(P, Q).$$

Further, for any $f : \mathcal{Z} \rightarrow [-1, 1]$, it holds that

$$|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq 4\sqrt{\mathbb{E}_Q[f^2] \cdot D_{\text{H}}^2(P, Q)} + 4D_{\text{H}}^2(P, Q).$$

Proof. We denote $P(z)$ (resp. $Q(z)$) to be the density function of P (resp. Q). Then for any function $f : \mathcal{Z} \rightarrow \mathbb{R}$,

$$\begin{aligned} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|^2 &= \left(\int_{\mathcal{Z}} f(z)(P(z) - Q(z))dz \right)^2 \\ &\leq \int_{\mathcal{Z}} f(z)^2 (\sqrt{P(z)} + \sqrt{Q(z)})^2 dz \cdot \int_{\mathcal{Z}} (\sqrt{P(z)} - \sqrt{Q(z)})^2 dz \\ &\leq 4D_{\text{H}}^2(P, Q) \cdot (\mathbb{E}_Q[f^2] + \mathbb{E}_P[f^2]). \end{aligned}$$

In particular, when $h : \mathcal{Z} \rightarrow [0, 1]$, the inequality above implies that

$$|\mathbb{E}_P[h] - \mathbb{E}_Q[h]| \leq 2D_{\text{H}}(P, Q) \sqrt{(\mathbb{E}_P[h] + \mathbb{E}_Q[h])} \leq \frac{1}{2}(\mathbb{E}_P[h] + \mathbb{E}_Q[h]) + 2D_{\text{H}}^2(P, Q),$$

and hence it holds that $\mathbb{E}_P[h] \leq 3\mathbb{E}_Q[h] + 4D_{\text{H}}^2(P, Q)$.

Now, we can bound

$$\begin{aligned} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|^2 &\leq 4D_{\text{H}}^2(P, Q) \cdot (\mathbb{E}_Q[f^2] + \mathbb{E}_P[f^2]) \\ &\leq 16D_{\text{H}}^2(P, Q) \cdot (\mathbb{E}_Q[f^2] + D_{\text{H}}^2(P, Q)). \end{aligned}$$

Combining the inequalities above with $D_{\text{H}}^2(\nu, \mu) \leq \frac{1}{2}D_{\text{KL}}(\nu \parallel \mu)$ completes the proof. $\square$

Lemma B.11. Suppose that $\hat{\mu} \propto \mu e^{-h}$, where $|h| \leq B$. Then for any distribution ν , it holds that

$$D_{\text{KL}}(\nu \parallel \hat{\mu}) \leq 4e^B D_{\text{KL}}(\nu \parallel \mu) + 4e^B \mathbb{E}_{\nu}[h^2].$$

Proof. By definition,

$$\begin{aligned} D_{\text{KL}}(\nu \parallel \hat{\mu}) - D_{\text{KL}}(\nu \parallel \mu) &= \mathbb{E}_{\nu}[\log \mu(x) - \log \hat{\mu}(x)] \\ &= \mathbb{E}_{\nu}[h] + \log \mathbb{E}_{\mu}[e^{-h}] \leq \mathbb{E}_{\nu}[h] - \mathbb{E}_{\mu}[h] + c_B \mathbb{E}_{\mu}[h^2], \end{aligned}$$

where $c_B = \frac{e^B - B - 1}{B^2}$. It holds that $\mathbb{E}_{\mu}[h^2] \leq 2\mathbb{E}_{\nu}[h^2] + 2B^2 D_{\text{H}}^2(\nu, \mu)$ because

$$\mathbb{E}_{\mu}[h^2] = \int h^2 d(\sqrt{\nu} + \sqrt{\mu} - \sqrt{\nu})^2 \leq 2 \int h^2 d\nu + 2 \int h^2 (\sqrt{\mu} - \sqrt{\nu})^2 \leq 2\mathbb{E}_{\nu}[h^2] + 4B^2 D_{\text{H}}^2(\nu, \mu).$$

Also,

$$\begin{aligned} |\mathbb{E}_{\nu}[h] - \mathbb{E}_{\mu}[h]| &\leq \sqrt{\int h^2 (\sqrt{\mu} + \sqrt{\nu})^2 \int (\sqrt{\mu} - \sqrt{\nu})^2} \leq \sqrt{4D_{\text{H}}^2(\nu, \mu) (\mathbb{E}_{\mu}[h^2] + \mathbb{E}_{\nu}[h^2])} \\ &\leq \sqrt{4D_{\text{H}}^2(\nu, \mu) (3\mathbb{E}_{\nu}[h^2] + 4B^2 D_{\text{H}}^2(\nu, \mu))} \leq \sqrt{12\mathbb{E}_{\nu}[h^2] D_{\text{H}}^2(\nu, \mu) + 4B D_{\text{H}}^2(\nu, \mu)} \\ &\leq 2\mathbb{E}_{\nu}[h^2] + (4B + 2)D_{\text{H}}^2(\nu, \mu). \end{aligned}$$

Combining the inequalities above with $D_{\text{H}}^2(\nu, \mu) \leq \frac{1}{2}D_{\text{KL}}(\nu \parallel \mu)$ completes the proof. $\square$

Lemma B.12. Suppose that $|f(x) - g(x)| \leq u(x) + v(x)$ such that $0 \leq u(x) \leq B$ for any x . Then

$$D_{\text{KL}}(\mu_f \parallel \mu_g) \leq e^B \mathbb{E}_{\mu_f}[u(x)^2] + (e^{v(x)} - 1)_+.$$

Proof. By definition,

$$D_{\text{KL}}(\mu_f \parallel \mu_g) = \mathbb{E}_{\mu_f}[g(x) - f(x)] + \log \mathbb{E}_{\mu_f} e^{f(x) - g(x)}.$$

Note that $e^w \leq 1 + w + \frac{e^B}{2}w^2$ for any $w \leq B$. Now, we can verify that

$$e^r \leq 1 + r + Cw^2 + (e^{|r|} - e^w)_+, \quad r \in \mathbb{R}, w \in [0, B],$$

where $C = \frac{e^B}{2}$. Specifically, when $|r| \leq w$, this reduces to $e^r \leq 1 + r + Cw^2$; when $r > w$, this reduces to $e^w \leq 1 + r + Cw^2$; and when $r < -w$, this reduces to $e^w - 1 - w - Cw^2 \leq e^{-r} - e^r + r - w$, and we can use $e^t - e^{-t} \geq 2t$ which holds for all $t \geq 0$. Applying this inequality gives

$$\begin{aligned} D_{\text{KL}}(\mu_f \parallel \mu_g) &\leq \mathbb{E}_{\mu_f}[g(x) - f(x)] + \mathbb{E}_{\mu_f} e^{f(x)-g(x)} - 1 \\ &\leq C \mathbb{E}_{\mu_f}[u(x)^2] + \mathbb{E}_{\mu_f}(e^{|f(x)-g(x)|} - e^{u(x)})_+ \\ &\leq e^B \mathbb{E}_{\mu_f}[u(x)^2 + (e^{v(x)} - 1)_+]. \end{aligned}$$

□

Lemma B.13. For any $\ell > 1$, it holds that

$$\bar{D}_\ell(\mu_f \parallel \mu_g) \leq \mathbb{E}_{x \sim \mu_f}[e^{2\ell|f(x)-g(x)|} - 1]. \quad (19)$$

Proof. By definition, we can write

$$\frac{\mu_f(x)}{\mu_g(x)} = e^{g(x)-f(x)} \mathbb{E}_{\mu_f}[e^{f-g}].$$

Therefore, we have

$$\begin{aligned} 1 + D_\ell(\mu_f \parallel \mu_g) &= \mathbb{E}_{\mu_f} \left(\frac{\mu_f}{\mu_g} \right)^{\ell-1} = (\mathbb{E}_{\mu_f}[e^{f-g}])^{\ell-1} \cdot \mathbb{E}_{\mu_f}[e^{(\ell-1)(g-f)}] \\ &\leq (\mathbb{E}_{\mu_f} e^{(\ell-1)|f-g|})^2 \leq \mathbb{E}_{\mu_f}[e^{2\ell|f-g|}]. \end{aligned}$$

Similarly,

$$\begin{aligned} 1 + D_\ell(\mu_g \parallel \mu_f) &= \mathbb{E}_{\mu_f} \left(\frac{\mu_g}{\mu_f} \right)^\ell = (\mathbb{E}_{\mu_f}[e^{f-g}])^{-\ell} \cdot \mathbb{E}_{\mu_f}[e^{\ell(f-g)}] \leq \mathbb{E}_{\mu_f}[e^{-\ell(f-g)}] \cdot \mathbb{E}_{\mu_f}[e^{\ell(f-g)}] \\ &\leq (\mathbb{E}_{\mu_f} e^{\ell|f-g|})^2 \leq \mathbb{E}_{\mu_f}[e^{2\ell|f-g|}]. \end{aligned}$$

Combining both inequalities completes the proof. □

Lemma B.14. Suppose that μ is a measure over $\mathbb{R}^d$ such that $\nabla \log \mu$ is L -Lipschitz. Then for any $\sigma > 0$, it holds that

$$\log(1 + D_{\chi^2}(\mu \parallel \mu * \mathbf{N}(0, \sigma^2 I))) \leq \frac{d}{2} \log(1 + L\sigma^2).$$

Proof. We denote $\nu = \mu * \mathbf{N}(0, \sigma^2 I)$, and then

$$\nu(x) = \int_{\mathbb{R}^d} \frac{1}{(2\pi\sigma^2)^{d/2}} e^{-\frac{\|z\|^2}{2\sigma^2}} \mu(x+z) dz.$$

Using the smoothness of $\log \mu$, we can bound

$$\log \mu(x+z) - \log \mu(x) - \langle \nabla \log \mu(x), z \rangle \geq -\frac{L}{2} \|z\|^2.$$

Hence, it holds that

$$\frac{\nu(x)}{\mu(x)} \geq \int_{\mathbb{R}^d} \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|z\|^2}{2} \left(L + \frac{1}{\sigma^2}\right) + \langle \nabla \log \mu(x), z \rangle\right) dz \geq (1 + \sigma^2 L)^{-d/2}.$$

This immediately implies the desired upper bound. □

The following lemma can be regarded as a “chain rule” of Hellinger distance (Jayram, 2009), see also Duchi (2023, Lemma 11.5.3) or Foster et al. (2024, Lemma D.2).

Lemma B.15 (Sub-additivity for squared Hellinger distance). *Suppose that $X_1 \rightarrow \dots \rightarrow X_T$ is a Markov chain. Given a family of transition kernels $\rho = (\rho_t : \mathcal{X} \rightarrow \Delta(\mathcal{X}))_{t \in [T]}$, we let $\mathbb{P}_\rho$ be the law of $X_1, \dots, X_T$ under $X_1 \sim \rho_1, X_t \sim \rho_t(\cdot | X_{t-1})$. Then, for any families of transition kernels ρ, ρ' , it holds that*

$$D_{\text{H}}^2(\mathbb{P}_\rho, \mathbb{P}_{\rho'}) \leq 7 \sum_{t=1}^T \mathbb{E}_{X_{t-1} \sim \mathbb{P}_\rho} [D_{\text{H}}^2(\rho_t(\cdot | X_{t-1}), \rho'_t(\cdot | X_{t-1}))],$$

where we regard $X_0 = \perp$ and $\rho_1(\cdot | \perp) = \rho_1$.

C. Proof of Theorem 3.1

Proof of Theorem 3.1. Let x_{out} denote the output of Algorithm 1. On a given iteration of the algorithm, conditional on the draw $x \sim q$, the probability that the algorithm terminates on that iteration is

$$\begin{aligned} a(x) &= \mathbb{E} \left[\prod_{j=1}^J \frac{B + W_j}{2B} \mid x \right] = \mathbb{E} \left[\left(\frac{B + \mathbb{E}[W_1 | x]}{2B} \right)^J \right] \\ &= \sum_{J \geq 0} \frac{e^{-2B} (2B)^J}{J!} \left(\frac{B + \mathbb{E}[W_1 | x]}{2B} \right)^J = e^{\mathbb{E}[W_1 | x] - B}. \end{aligned}$$

Thus, the acceptance probability of a given iteration is $A := \int a(x) q(dx)$. Then,

$$\begin{aligned} \mathbb{P}(x_{\text{out}} \in dx) &= \sum_{i=1}^{\infty} \mathbb{P}(x_{\text{out}} \in dx, \text{Algorithm 1 succeeds on iteration } i) \\ &= \sum_{i=1}^{\infty} (1 - A)^{i-1} q(dx) a(x) = \frac{q(dx) a(x)}{A}. \end{aligned}$$

For the second statement, let $J_1, J_2, J_3, \dots$ be an i.i.d. sequence of $\text{Poisson}(2B)$ random variables. The probability that the number N of draws from $\mathcal{W}$ exceeds m is $\mathbb{P}(N \geq m) = \mathbb{P}(\sum_{i=1}^I J_i \geq m)$, where I is the number of iterations of the algorithm. For any i_0 , we take $m = (2 + c) i_0 B$ and bound

$$\mathbb{P} \left(\sum_{i=1}^I J_i \geq m \right) \leq \mathbb{P}(I > i_0) + \mathbb{P} \left(\sum_{i=1}^{i_0} J_i \geq (2 + c) i_0 B \right).$$

Note that if any $J_i = 0$, then $I \leq i$, so the first term is bounded by $(1 - e^{-2B})^{i_0} \leq \exp(-e^{-2B} i_0)$. The second term, using a concentration bound for the Poisson random variable, is bounded by $\exp(-\frac{c^2}{c+2} i_0 B)$. Setting $i_0 = e^{2B} \log(2/\delta)$ and $c = e^{-2B}$ suffices to bound both terms by $\delta/2$.

Note that this also implies $\mathbb{E}[\exp(N/(CBe^{2B}))] \leq 2$ for an absolute constant $C > 0$, where $N = \sum_{i=1}^I J_i$ is the total number of draw from $\mathcal{W}$. So, the last statement follows immediately. $\square$

D. Proofs for Section 3

In the following, we prove a slightly stronger version of Theorem 3.3.

Theorem D.1. *Suppose that Assumption 3.2 holds, and Algorithm 1 is instantiated as in Theorem 3.3. Let $B = \Theta(1)$, $\ell \geq 2$, $\delta \in (0, \frac{1}{2}]$, and*

$$\frac{1}{\eta^{1+s}} \gg \beta_s^2 \left(d^s (\ell + \log(1/\delta)) + \frac{s}{d^{1-s}} (\ell^2 + \log^2(1/\delta)) \right).$$

Then the law $\hat{\nu}$ of Algorithm 1 satisfies

$$\max\{D_\ell(\nu \| \hat{\nu}), D_\ell(\hat{\nu} \| \nu)\} \leq \delta.$$

Proof of Theorem D.1. We recall that $q = \mathcal{N}(\hat{x}, \eta I)$, where we denote $\hat{x} := x_0 - \eta \nabla f(x_+)$.

By Theorem 3.1, the output of Algorithm 1 with the specified choices samples from $\hat{\nu}$, such that

$$\log \hat{\nu}(x) - \log q(x) = \text{const} + \mathbb{E}_{z \sim P, r \sim \text{Unif}([0,1])} \text{Clip}_B(\langle \dot{\gamma}_{z,r}(x), \nabla f(x_+) - \nabla f(\gamma_{z,r}(x)) \rangle).$$

In the following, we denote $\mathbb{E}_{z,r}[\cdot] := \mathbb{E}_{z \sim P, r \sim \text{Unif}([0,1])}[\cdot]$ and define

$$W_{r,z,x} := \langle \dot{\gamma}_{z,r}(x), \nabla f(x_+) - \nabla f(\gamma_{z,r}(x)) \rangle.$$

Then, $|\log \nu(x) - \log \hat{\nu}(x) - \text{const}| \leq V(x)$, where

$$V(x) = \mathbb{E}_{z,r} \tau_B(W_{r,z,x}),$$

where we recall $\tau_B(y) := (|y| - B)_+ = |y - \text{Clip}_B(y)|$. Then, using Lemma B.13, we can bound

$$\begin{aligned} \max\{D_\ell(\nu \parallel \hat{\nu}), D_\ell(\hat{\nu} \parallel \nu)\} &\leq \mathbb{E}_{x \sim \hat{\nu}}[e^{2\ell V(x)} - 1] \leq e^{2B} \mathbb{E}_{x \sim q}[e^{2\ell V(x)} - 1] \\ &\leq e^{2B} \mathbb{E}_{x \sim q} \mathbb{E}_{z,r}[e^{2\ell \tau_B(W_{r,z,x})} - 1], \end{aligned} \quad (20)$$

where we use $\frac{d\hat{\nu}}{dq}(x) \leq e^{2B}$ for $x \in \mathbb{R}^d$ and the convexity of $w \mapsto e^w$.

In the following, we proceed to prove the following claims.

Claim 1. It holds that for any fixed $r \in [0, 1]$ and $\lambda^2 \leq \frac{d^{1-s}}{12s\beta_s^2\eta^{1+s}}$,

$$\log \mathbb{E}_{x \sim q, z \sim P} \exp(\lambda |W_{r,z,x}|) \leq 10d^s \eta^{1+s} \lambda^2 \beta_s^2.$$

Proof of Claim 1. Recall that $a_r = \sin(\pi r/2)$, $b_r = \cos(\pi r/2)$,

$$\gamma_{z,r}(x) = \hat{x} + a_r(x - \hat{x}) + b_r z, \quad \dot{\gamma}_{z,r}(x) = a'_r(x - \hat{x}) + b'_r z.$$

Hence, under $x \sim q = \mathcal{N}(\hat{x}, \eta I)$ and $z \sim P = \mathcal{N}(0, \eta I)$, $[\gamma_{z,r}(x); \dot{\gamma}_{z,r}(x)]$ are jointly distributed as

$$[\gamma_{z,r}(x); \dot{\gamma}_{z,r}(x)] \sim \mathcal{N}\left(\begin{bmatrix} \hat{x} \\ 0 \end{bmatrix}, \begin{bmatrix} \eta I & \\ & (\pi/2)^2 \eta I \end{bmatrix}\right).$$

Hence, as long as $3\eta\lambda^2\beta_s^2 \leq \frac{d^{1-s}}{4s\eta^s}$,

$$\begin{aligned} \mathbb{E}_{x \sim q, z \sim P} \exp(\lambda |W_{r,z,x}|) &= \mathbb{E}_{Z_1, Z_2 \sim \mathcal{N}(0, \eta I)} \exp(\lambda |\langle Z_1 \pi/2, \nabla f(\hat{x} + Z_2) - \nabla f(x_+) \rangle|) \\ &\leq 2 \mathbb{E}_{Z_2 \sim \mathcal{N}(0, \eta I)} \exp\left(\frac{3}{2} \eta \lambda^2 \|\nabla f(\hat{x} + Z_2) - \nabla f(x_+)\|^2\right) \\ &\leq 2 \mathbb{E}_{Z_2 \sim \mathcal{N}(0, \eta I)} \exp\left(\frac{3}{2} \eta \lambda^2 \beta_s^2 \|\hat{x} - x_+ + Z_2\|^{2s}\right) \\ &\leq 2 \exp(3\eta\lambda^2\beta_s^2 \|\hat{x} - x_+\|^{2s}) \mathbb{E}_{W \sim \mathcal{N}(0, \eta I)} \exp(3\eta\lambda^2\beta_s^2 \|W\|^{2s}) \\ &\leq 2 \exp(3\eta\lambda^2\beta_s^2 \|\hat{x} - x_+\|^{2s} + 6d^s \eta^{1+s} \lambda^2 \beta_s^2) \leq 2 \exp(10d^s \eta^{1+s} \lambda^2 \beta_s^2), \end{aligned}$$

where the second line uses Lemma B.5, and the last line uses Lemma B.6. This completes the proof of Claim 1.

Finally, we use Claim 1 to prove the following claim, from which Theorem D.1 follows immediately.

Claim 2. Suppose that

$$\frac{1}{\eta^{1+s}} \geq 64\beta_s^2 \left(\ell B^{-1} d^s + \frac{s\ell^2}{d^{1-s}} \right).$$

Then for any $r \in [0, 1]$, it holds that

$$\mathbb{E}_{x \sim q, z \sim P} [e^{2\ell \tau_B(W_{r,z,x})} - 1] \leq 2 \exp\left(-\min\left\{\frac{B^2}{40\beta_s^2 d^s \eta^{1+s}}, \frac{Bd^{(1-s)/2}}{8\sqrt{s}\beta_s \eta^{(1+s)/2}}\right\}\right).$$

Proof of Claim 2. Using Claim 1, for any fixed $r \in [0, 1]$ and $2\ell \leq \lambda \leq \frac{d^{(1-s)/2}}{4\sqrt{s}\beta_s\eta^{(1+s)/2}}$, we can upper bound

$$\begin{aligned} \mathbb{E}_{x \sim q, z \sim P}[e^{2\ell\tau_B(W_{r,z,x})} - 1] &\leq \mathbb{E}_{x \sim q, z \sim P}[e^{\lambda\tau_B(W_{r,z,x})} - 1] \leq e^{-\lambda B} \mathbb{E}_{x \sim q, z \sim P}[e^{\lambda|W_{r,z,x}|}] \\ &\leq 2 \exp(10d^s\eta^{1+s}\lambda^2\beta_s^2 - B\lambda). \end{aligned}$$

Therefore, the desired upper bound follows from setting

$$\lambda = \min\left\{\frac{d^{(1-s)/2}}{4\sqrt{s}\beta_s\eta^{(1+s)/2}}, \frac{B}{20\beta_s^2d^s\eta^{1+s}}\right\} \geq 2\ell.$$

Combining Claim 2 above with Eq. (20), we can deduce the desired upper bound. $\square$

E. Proofs for Section 4

E.1. Proof of Theorem 4.1

Proof of Theorem 4.1. The process $X_T \rightarrow \dots \rightarrow X_1$ generated by Algorithm 2 is a Markov chain such that $X_T \sim \hat{p}_T$, and $X_t | X_{t+1} \sim \hat{\rho}_t(\cdot | X_{t+1})$. The chain rule for the KL divergence yields

$$D_{\text{KL}}(p_1 \| \hat{p}_1) = D_{\text{KL}}(p_T \| \hat{p}_T) + \sum_{t=1}^{T-1} \mathbb{E}_{X_{t+1} \sim p_{t+1}} D_{\text{KL}}(\rho_t(\cdot | X_{t+1}) \| \hat{\rho}_t(\cdot | X_{t+1})).$$

Given $X_{t+1} = x_{t+1}$ and $\bar{x}_t = \alpha_t^{-1}x_{t+1}$, Algorithm 1 is instantiated with base measure $q_t = \mathcal{N}(\bar{x}_t, \eta_t I)$. Consider the target $\rho_t(\cdot | x_{t+1})$, and by Theorem 3.1,

$$\begin{aligned} &\log \hat{\rho}_t(x | x_{t+1}) - \log \rho_t(x | x_{t+1}) - \text{const} \\ &= \mathbb{E}_{r \sim \text{Unif}([0,1])} \text{Clip}_B(\langle x - \bar{x}_t, \mathbf{s}_t(rx + (1-r)\bar{x}_t) \rangle) - \log p_t(x) \\ &= \mathbb{E}_{r \sim \text{Unif}([0,1])} [\text{Clip}_B(\langle x - \bar{x}_t, \mathbf{s}_t(rx + (1-r)\bar{x}_t) \rangle) - \langle x - \bar{x}_t, \nabla \log p_t(rx + (1-r)\bar{x}_t) \rangle], \end{aligned}$$

and hence

$$\begin{aligned} |\log \hat{\rho}_t(x | x_{t+1}) - \log \rho_t(x | x_{t+1}) - \text{const}| &\leq \underbrace{\mathbb{E}_{r \sim \text{Unif}([0,1])} \text{Clip}_B(|\langle x - \bar{x}_t, [\mathbf{s}_t - \nabla \log p_t](rx + (1-r)\bar{x}_t) \rangle|)}_{=: u(x)} \\ &\quad + \underbrace{\mathbb{E}_{r \sim \text{Unif}([0,1])} \tau_B(\langle x - \bar{x}_t, \nabla \log p_t(rx + (1-r)\bar{x}_t) \rangle)}_{=: v(x)}, \end{aligned}$$

where we used $|a - \text{Clip}_B(b)| \leq \text{Clip}_B(|a - b|) + \tau_B(b)$ for $a, b \in \mathbb{R}$. Applying Lemma B.12 gives

$$\begin{aligned} D_{\text{KL}}(\rho_t(\cdot | x_{t+1}) \| \hat{\rho}_t(\cdot | x_{t+1})) &\leq e^B \mathbb{E}_{x \sim \rho_t(\cdot | \bar{x}_t), r \sim \text{Unif}([0,1])} [\langle x - \bar{x}_t, [\nabla \log p_t - \mathbf{s}_t](rx + (1-r)\bar{x}_t) \rangle^2 \wedge B^2] \\ &\quad + \mathbb{E}_{x \sim \rho_t(\cdot | \bar{x}_t), r \sim \text{Unif}([0,1])} (\exp(\langle x - x', \nabla \log p_t(rx + (1-r)\bar{x}_t) \rangle) - e^B)_+. \end{aligned}$$

Taking expectation over $X_{t+1} \sim p_{t+1}$ and using the fact that $X_{t+1} | X_t \sim \mathcal{N}(\alpha_t X_t, \alpha_t^2 \eta_t I)$, we can conclude that

$$\begin{aligned} &\mathbb{E}_{X_{t+1} \sim p_{t+1}} D_{\text{KL}}(\rho_t(\cdot | X_{t+1}) \| \hat{\rho}_t(\cdot | X_{t+1})) \\ &\leq e^B \mathbb{E}_{X_t \sim p_t, r \sim \text{Unif}([0,1]), Z \sim \mathcal{N}(0, \eta_t I)} [\langle Z, [\nabla \log p_t - \mathbf{s}_t](X_t + rZ) \rangle^2 \wedge B^2] \\ &\quad + \mathbb{E}_{X_t \sim p_t, r \sim \text{Unif}([0,1]), Z \sim \mathcal{N}(0, \eta_t I)} (\exp(|\langle Z, \nabla \log p_t(X_t + rZ) \rangle|) - e^B)_+. \end{aligned}$$

By Proposition E.1 (1), it holds that for any $r \in [0, 1]$,

$$\begin{aligned} &\mathbb{E}_{X_t \sim p_t, Z \sim \mathcal{N}(0, \eta_t I)} [\langle Z, [\nabla \log p_t - \mathbf{s}_t](X_t + rZ) \rangle^2 \wedge B^2] \\ &\leq 2 \mathbb{E}_{X_t \sim p_t, Z \sim \mathcal{N}(0, \eta_t I)} [\langle Z, [\nabla \log p_t - \mathbf{s}_t](X_t) \rangle^2 \wedge B^2] + O(B^2) \cdot e^{-K/12} \\ &\leq 2\eta_t \mathbb{E}_{X_t \sim p_t} \|\nabla \log p_t(X_t) - \mathbf{s}_t(X_t)\|^2 + O(B^2) \cdot e^{-K/12}, \end{aligned}$$

where we use $\mathbb{E}_{Z \sim \mathcal{N}(0, \eta_t I)}[ZZ^\top] = \eta_t I$.

By [Proposition E.1](#) (2) and the fact that $\mathbb{E}[Y \mathbb{I}\{Y > M\}] \leq M^{\lambda-1} \mathbb{E}[Y^\lambda]$ for $\lambda > 1$, it holds that for any $r \in [0, 1]$,

$$\mathbb{E}_{X_t \sim p_t, Z \sim \mathcal{N}(0, \eta_t I)}(\exp(|\langle Z, \nabla \log p_t(X_t + rZ) \rangle|) - e^B)_+ \leq O(e^B) \cdot e^{-c_B K}.$$

Combining these inequalities completes the proof. $\square$

Proposition E.1. Suppose that $\eta \leq \frac{\sigma^2}{\max\{Kd, K^2\}}$ and $K \geq 2$, and $P = \bar{P} * \mathcal{N}(0, \sigma^2 I)$.

(1) For any $0 \leq F \leq 1$ and $r \in [0, 1]$, it holds that

$$\mathbb{E}_{X \sim P, Z \sim \mathcal{N}(0, \eta I)} F(Z, X + rZ) - 2 \mathbb{E}_{X \sim P, Z \sim \mathcal{N}(0, \eta I)} F(Z, X) \leq 2 \exp\left(-\frac{K}{12}\right).$$

(2) For any $r \in [0, 1]$ and $B > 0$, it holds that

$$\mathbb{E}_{X \sim P, Z \sim \mathcal{N}(0, \eta I)}(\exp(|\langle Z, \nabla \log p(X + rZ) \rangle|) - e^B)_+ \leq e^{B+1} \cdot \exp\left(-\frac{\min\{B, 1\}B}{32}K\right).$$

Proof. (1) Fix any X_0 , and let $Q_{X_0}^r$ be the distribution of (W, Z) under $X \sim \mathcal{N}(X_0, \sigma^2 I)$, $Z \sim \mathcal{N}(0, \eta I)$, and $W = X + rZ$. In other words, $Q_{X_0}^r = \mathcal{N}([X_0; \mathbf{0}], \Sigma_r)$ where $\Sigma_r = \begin{bmatrix} (\sigma^2 + r^2\eta)I & r\eta I \\ r\eta I & \eta I \end{bmatrix}$. We consider the λ -divergence $D_\lambda(p \| q) := \mathbb{E}_q\left(\frac{dp}{dq}\right)^\lambda - 1$. Then, as long as $\lambda(\lambda - 1)r^2\eta < \sigma^2$, we can calculate

$$\begin{aligned} 2 \log(1 + D_\lambda(Q_{X_0}^r \| Q_{X_0}^0)) &= (\lambda - 1) \log \det(\Sigma_0) - \lambda \log \det(\Sigma_r) - \log \det(\lambda \Sigma_r^{-1} - (\lambda - 1) \Sigma_0^{-1}) \\ &= (\lambda - 1) \log \det(\Sigma_0) - (\lambda - 1) \log \det(\Sigma_r) - \log \det(\lambda I - (\lambda - 1) \Sigma_r \Sigma_0^{-1}) \\ &= -d \log\left(1 - \frac{\lambda(\lambda - 1)r^2\eta}{\sigma^2}\right). \end{aligned}$$

In particular, as long as $\lambda^2 \leq \frac{\sigma^2}{2\eta}$, it holds that

$$1 + \max_{r \in [0, 1]} D_\lambda(Q_{X_0}^r \| Q_{X_0}^0) \leq \exp\left(\frac{d\lambda^2\eta}{\sigma^2}\right).$$

Now, consider the following distribution of (X_0, W, Z) :

$$\mathbf{Q}^r : X_0 \sim \bar{P}, X \sim \mathcal{N}(X_0, \sigma^2 I), Z \sim \mathcal{N}(0, \eta I), W = X + rZ.$$

By the data-processing inequality, we know

$$1 + D_\lambda(\mathbf{Q}^r \| \mathbf{Q}^0) \leq \exp\left(\frac{d\lambda^2\eta}{\sigma^2}\right) \leq \exp\left(\frac{\lambda^2}{K}\right), \quad \text{for all } 1 \leq \lambda \leq \frac{\sigma}{\sqrt{2\eta}}, \quad r \in [0, 1].$$

In particular, choosing $\lambda = \frac{K}{2}$ gives $2^{-\lambda} (1 + D_\lambda(\mathbf{Q}^r \| \mathbf{Q}^0)) \leq \exp(-\frac{K}{12})$. Applying [Lemma B.7](#) completes the proof.

(2) We denote $f = \log P$. By [Proposition E.8](#), for $0 \leq \lambda \leq \frac{\sigma^2}{4}$, we can bound

$$\mathbb{E}_{X \sim P} \exp(\lambda \|\nabla f(X)\|^2) \leq \exp\left(\frac{2d\lambda}{\sigma^2}\right).$$

Next, recalling the definition of $\mathbf{Q}^r$, we can write

$$\begin{aligned} \mathbb{E}_{X \sim P, Z \sim \mathcal{N}(0, \eta I)} \exp(\lambda |\langle Z, \nabla f(X + rZ) \rangle|) &= \mathbb{E}_{X_0 \sim \bar{P}} \mathbb{E}_{X \sim \mathcal{N}(X_0, \sigma^2 I), Z \sim \mathcal{N}(0, \eta I)} \exp(\lambda |\langle Z, \nabla f(X + rZ) \rangle|) \\ &= \mathbb{E}_{(Z, W) \sim \mathbf{Q}^r} \exp(\lambda |\langle Z, \nabla f(W) \rangle|). \end{aligned}$$

By the Cauchy–Schwarz inequality,

$$\begin{aligned}\mathbb{E}_{(W,Z) \sim \mathbf{Q}^r} \exp(\lambda |\langle Z, \nabla f(W) \rangle|) &\leq \sqrt{(1 + D_{\chi^2}(\mathbf{Q}^r \parallel \mathbf{Q}^0)) \mathbb{E}_{(W,Z) \sim \mathbf{Q}^0} \exp(2\lambda |\langle Z, \nabla f(W) \rangle|)} \\ &\leq e \cdot \sqrt{\mathbb{E}_{W \sim \mathbf{Q}^0} [e^{4\lambda^2 \eta \|\nabla f(W)\|^2}]},\end{aligned}$$

where we use that under $\mathbf{Q}^0$, $\langle Z, \nabla f(W) \rangle \mid W \sim \mathcal{N}(0, \eta \|\nabla f(W)\|^2)$. Note that the distribution of $W \sim \mathbf{Q}^0$ is exactly P . Hence, this implies that as long as $8\lambda^2 \eta \leq \frac{\sigma^2}{4}$,

$$\mathbb{E}_{(W,Z) \sim \mathbf{Q}^r} \exp(\lambda |\langle Z, \nabla f(W) \rangle|) \leq \exp\left(1 + \frac{8d\lambda^2 \eta}{\sigma^2}\right) \leq \exp\left(1 + \frac{8\lambda^2}{K}\right).$$

Taking $\lambda = \frac{\min\{B, 1\}}{16} K$ gives

$$\begin{aligned}\mathbb{E}_{(W,Z) \sim \mathbf{Q}^r} (\exp(|\langle Z, \nabla f(W) \rangle|) - e^B)_+ &\leq e^{-(\lambda-1)B} \mathbb{E}_{(W,Z) \sim \mathbf{Q}^r} \exp(\lambda |\langle Z, \nabla f(W) \rangle|) \\ &\leq \exp\left(B + 1 - \frac{\min\{B, 1\}B}{32} K\right).\end{aligned}$$

□

E.2. Analysis of a DDPM-Like Method

In this subsection, our goal is to sample from the tilt measure

$$\nu_{x_0}(x) \propto p(x) \exp\left(-\frac{\|x - x_0\|^2}{2\eta}\right), \quad (21)$$

given an estimated score function $s \approx \nabla \log p$. We carry out our analysis under the following assumption, which will hold true in the diffusion model setting.

Assumption E.2. The distribution p takes the form $p = \bar{p} * \mathcal{N}(0, \sigma^2 I)$. We write $\check{p} = p * \mathcal{N}(0, \eta I)$.

For the diffusion model applications, we take $p = p_t = \bar{p}_t * \mathcal{N}(0, \sigma^2 I)$, where $\bar{p}_t$ is the distribution of $\bar{\alpha}_t X_0$, $X_0 \sim p_{\text{data}}$. Further, under the true diffusion process, we have $\alpha_t^{-1} X_{t+1} \sim p_t * \mathcal{N}(0, \eta_t I)$.

Notation. We decompose $\frac{1}{\eta} = \frac{1}{\bar{\eta}} - \lambda$ for some $\lambda \geq 0$. For [Theorem 4.4](#), we can take the simple choice $\lambda = 0$, but our proof of [Theorem 4.7](#) requires taking $\lambda > 0$. To unify the analyses of the two results, we consider a general choice of λ here.

We fix the parameter $\lambda \geq 0$, and we denote $s^* = \nabla \log p$,

$$\tilde{s}(x) = s(x) + \lambda x, \quad \tilde{s}^*(x) = s^*(x) + \lambda x.$$

We also write $\bar{B} = \max\{B, B^{-1}\}$.

In the following, we abbreviate $\mathbb{E}_{r,z}[\cdot] := \mathbb{E}_{r \sim \text{Unif}([0,1]), z \sim \mathcal{N}(0, \eta I)}[\cdot]$.

Distributions. By [Theorem 3.1](#), [Algorithm 1](#) instantiated with the choices $q = \mathcal{N}(x_0 + \bar{\eta}s(x_0), \bar{\eta}I)$ and

$$\widehat{W}_{z,r,x} = \text{Clip}_B(\langle \dot{\gamma}_{z,r,x_0}(x), \tilde{s}(\gamma_{z,r,x_0}(x)) - \tilde{s}(x_0) \rangle) \quad (22)$$

samples from the distribution $\hat{\rho}_{x_0}(\cdot)$, where

$$\log \hat{\rho}_{x_0}(x) = \text{const} + \mathbb{E}_{r,z} \text{Clip}_B(\langle \dot{\gamma}_{z,r,x_0}(x), \tilde{s}(\gamma_{z,r,x_0}(x)) - \tilde{s}(x_0) \rangle) - \frac{\|x - x_0 - \bar{\eta}s(x_0)\|^2}{2\bar{\eta}}. \quad (23)$$

We write $\rho_{x_0}^*(\cdot)$ for the density corresponding to score $s = s^*$, i.e.,

$$\log \rho_{x_0}^*(x) = \text{const} + \mathbb{E}_{r,z} \text{Clip}_B(\langle \dot{\gamma}_{z,r,x_0}(x), \tilde{s}^*(\gamma_{z,r,x_0}(x)) - \tilde{s}^*(x_0) \rangle) - \frac{\|x - x_0 - \bar{\eta}s^*(x_0)\|^2}{2\bar{\eta}}. \quad (24)$$

Using Eq. (8), we can also write

$$\begin{aligned} \log \nu_{x_0}(x) + \frac{\|x - x_0 - \bar{\eta} \mathbf{s}^*(x_0)\|^2}{2\bar{\eta}} &= \text{const} + \log p(x) - \langle \mathbf{s}^*(x_0), x - x_0 \rangle + \frac{\lambda}{2} \|x - x_0\|^2 \\ &= \text{const} + \mathbb{E}_{r,z} \langle \dot{\gamma}_{z,r,x_0}(x), \tilde{\mathbf{s}}^*(\gamma_{z,r,x_0}(x)) - \tilde{\mathbf{s}}^*(x_0) \rangle, \end{aligned}$$

where we recall $\mathbf{s}^* = \nabla \log p$.

Proposition E.3 (Convergence under true score). *Suppose that Assumption E.2 holds and $\mathbf{s} = \mathbf{s}^*$, $B = \Theta(1)$, $G \geq K \geq 1$, $\eta \leq \frac{\sigma^2}{4BG}$. We define the set*

$$\begin{aligned} \mathcal{E}_{G,K} = \left\{ (\bar{x}, V) : H = \nabla \tilde{\mathbf{s}}^*(\bar{x} + V), \|H\|_{\text{op}} \leq \frac{G}{K\sigma^2}, \|H\|_{\text{F}} \leq \frac{G}{\sqrt{K}\sigma^2}, \right. \\ \left. |\text{tr}(H)| \leq \frac{G^2}{\sigma^2}, |V^\top H V| \leq G^2, \|HV\|^2 \leq \frac{G^3}{K\sigma^2} \right\}. \end{aligned}$$

Let $\rho_{x_0}^*(\cdot)$ be the probability density defined in (24). Then it holds that

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim \frac{d^{4.5}\eta}{\sigma^2} \left(e^{-cK} + \sqrt{\max_{\eta' \in [0, 2\eta]} \mathbb{P}_{\bar{x} \sim \bar{p}, V \sim \mathcal{N}(0, (\sigma^2 + \eta')I)}((\bar{x}, V) \notin \mathcal{E}_{G,K})} \right).$$

Proposition E.4 (Score error). *Fix any $\delta \in (0, 1)$, $\eta > 0$. Suppose that Assumption E.2 holds.*

Let $\hat{\rho}_{x_0}(\cdot)$ be the probability density of x defined in (23) with score function $\mathbf{s}$. Then, the following holds.

(1) *It holds that*

$$\begin{aligned} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \hat{\rho}_{x_0}(\cdot)) &\lesssim e^{4B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \\ &\quad + e^{4B} \bar{B}^2 \left(\delta + G \max_{\eta' \in [0, 2\eta]} \mathbb{E}_{X \sim p * \mathcal{N}(0, \eta'I)} \|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2 \right), \end{aligned}$$

where we denote $G = \eta + \frac{\eta^2}{\sigma^2} (d + \log(1/\delta))$.

(2) *Suppose that $\eta \leq \frac{\sigma^2}{8 \max\{\sqrt{d \log(1/\delta)}, \log(1/\delta)\}}$. Then, it holds that*

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{H}}^2(\nu_{x_0}(\cdot), \hat{\rho}_{x_0}(\cdot)) \lesssim e^{4B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + e^{4B} \bar{B}^2 (\delta + G \mathbb{E}_{X \sim p} \|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2).$$

We give the proofs of Proposition E.3 and Proposition E.4 in Appendix E.6 and Appendix E.7 respectively.

E.3. Proof of Theorem 4.4

As a direct corollary of Proposition E.3, we have the following result.

Corollary E.5. *Suppose that Assumption E.2 holds, $B = \Theta(1)$, $L \geq 1$, $\delta \in (0, \frac{1}{2})$, $\lambda = 0$, and*

$$\frac{\sigma^2}{\eta} \geq C_0 \max\{L \log(1/\delta), \sqrt{dL \log(1/\delta)}\},$$

where C_0 is a sufficiently large constant.

Suppose that given x_0 , Algorithm 1 is instantiated with (22), and let $\hat{\rho}_{x_0}(\cdot)$ be the output distribution (given input x_0). Then it holds that

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{H}}^2(\nu_{x_0}(\cdot), \hat{\rho}_{x_0}(\cdot)) \lesssim d^4 \left(\delta + \sqrt{\mathbb{P}_{x \sim p}(\|\nabla \mathbf{s}^*(x)\|_{\text{op}} \geq \frac{L}{\sigma^2})} \right) + G \mathbb{E}_{x \sim p} \|\mathbf{s}(x) - \mathbf{s}^*(x)\|^2,$$

where $\nu_{x_0}(x) \propto p(x) \exp\left(-\frac{\|x - x_0\|^2}{2\eta}\right)$ is the tilted measure defined in Eq. (21), and $G = \eta + \frac{\eta^2}{\sigma^2} (d + \log(1/\delta))$.

Proof of Corollary E.5. Fix any $\bar{\sigma}^2 \in [\sigma^2, \sigma^2 + 2\eta]$ and $K = \max\{c^{-1} \log(1/\delta), 1\}$. We write $\mathbb{P}_{\bar{\sigma}}(\cdot)$ to be the probability distribution of $(\bar{x}, V)$ under $\bar{x} \sim \bar{p}$ and $V \sim \mathcal{N}(0, \bar{\sigma}^2 I)$. Define

$$E_0 := \left\{ (\bar{x}, V) : \|\nabla \mathbf{s}^*(\bar{x} + V)\|_{\text{op}} \leq \frac{L}{\sigma^2} \right\}.$$

By Proposition E.8 in Appendix E.5 and Lemma B.5, it holds that $\mathbb{P}_{\bar{\sigma}}((\bar{x}, V) \notin E_1) \leq \delta^2$, where

$$E_1 := \left\{ (\bar{x}, V) : \|V\|^2 \leq C\sigma^2(d + \log(1/\delta)), \|\nabla \mathbf{s}^*(\bar{x} + V)\|_* \leq \frac{C(d + \log(1/\delta))}{\sigma^2} \right\}.$$

Now, suppose that

$$G \geq \max\{KL, \sqrt{C(d + \log(1/\delta))KL}\}.$$

We argue that $E_0 \cap E_1 \subseteq \mathcal{E}_{G,K}$. This is because for $(\bar{x}, V) \in E_0 \cap E_1$ and $H = \nabla \mathbf{s}^*(\bar{x} + V)$, we can bound

$$\begin{aligned} \|H\|_{\text{F}} &\leq \sqrt{\|H\|_{\text{op}} \cdot \|H\|_*} \leq \frac{\sqrt{C(d + \log(1/\delta))L}}{\sigma^2} \leq \frac{G}{\sqrt{K}\sigma^2}, \\ |\text{tr}(H)| &\leq \|H\|_* \leq \frac{C(d + \log(1/\delta))}{\sigma^2} \leq \frac{G^2}{\sigma^2}, \\ |V^\top HV| &\leq \|H\|_{\text{op}} \|V\|^2 \leq CL(d + \log(1/\delta)) \leq G^2, \\ \|HV\|^2 &\leq \|H\|_{\text{op}}^2 \|V\|^2 \leq \frac{CL^2(d + \log(1/\delta))}{\sigma^2} \leq \frac{G^3}{K\sigma^2}. \end{aligned}$$

This immediately gives $E_0 \cap E_1 \subseteq \mathcal{E}_{G,K}$, and hence

$$\mathbb{P}_{\bar{\sigma}}((\bar{x}, V) \notin \mathcal{E}_{G,K}) \leq \mathbb{P}_{\bar{\sigma}}(E_0^c) + \mathbb{P}_{\bar{\sigma}}(E_1^c).$$

Note that $\mathbb{P}_{\bar{\sigma}}(E_1^c) \leq \delta^2$ and by Proposition E.8,

$$\mathbb{P}_{\bar{\sigma}}(E_0^c) \leq e\mathbb{P}_{\sigma}(E_0^c) + e\delta^2.$$

Then, by Proposition E.3, it holds that

$$\mathbb{E}_{x_0 \sim \bar{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim d^4 \left(\delta + \sqrt{\mathbb{P}_{x \sim p}(\|\nabla \mathbf{s}^*(x)\|_{\text{op}} \geq \frac{L}{\sigma^2})} \right),$$

where $\rho_{x_0}^*(\cdot)$ is defined in Eq. (24). Finally, applying Proposition E.4 (2) gives the desired result. $\square$

Proof of Theorem 4.4. To prove Theorem 4.4, we recall that the process $X_T \rightarrow \dots \rightarrow X_1$ generated by Algorithm 2 is a Markov chain such that $X_T \sim \hat{p}_T$, and $X_t \mid X_{t+1} \sim \hat{\rho}_t(\cdot \mid X_{t+1})$, where $\hat{\rho}_t(\cdot \mid X_{t+1})$ is the conditional distribution of X_t given X_{t+1} . The chain rule for the Hellinger distance (Lemma B.15) yields

$$D_{\text{H}}^2(p_1, \hat{p}_1) \leq 7D_{\text{H}}^2(p_T, \hat{p}_T) + 7 \sum_{t=1}^{T-1} \mathbb{E}_{X_{t+1} \sim p_{t+1}} D_{\text{H}}^2(\rho_t(\cdot \mid X_{t+1}), \hat{\rho}_t(\cdot \mid X_{t+1})).$$

Note that under $X_{t+1} \sim p_{t+1}$, it holds that $\bar{x}_t = \alpha_t^{-1} X_{t+1} \sim p * \mathcal{N}(0, \eta_t I)$. Hence, Corollary E.5 implies that for $t \in [T]$, it holds that

$$\mathbb{E}_{X_{t+1} \sim p_{t+1}} D_{\text{H}}^2(\rho_t(\cdot \mid X_{t+1}), \hat{\rho}_t(\cdot \mid X_{t+1})) \lesssim \delta + \left(\eta + \frac{\eta^2}{\sigma^2} (d + \log(1/\delta)) \right) \mathbb{E}_{X \sim p_t} \|\mathbf{s}_t(X) - \mathbf{s}_t^*(X)\|^2.$$

The proof of Theorem 4.4 is then completed by combining inequalities above. $\square$

E.4. Proof of Theorem 4.7

As a direct corollary of Proposition E.3 and Proposition E.7, we have the following result.

Corollary E.6. Suppose that Assumption E.2 holds, $B = \Theta(1)$, $\delta \in (0, \frac{1}{2})$, $\lambda = \frac{1}{\sigma^2}$, and

$$\frac{\sigma^2}{\eta} \geq C_0 (d_* \log(d/\delta) + \log^2(d/\delta)),$$

where C_0 is a sufficiently large constant.

Suppose that given x_0 , Algorithm 1 is instantiated with (22), and let $\hat{\rho}_{x_0}(\cdot)$ be the output distribution (given input x_0). Then it holds that

$$\mathbb{E}_{x_0 \sim \tilde{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \hat{\rho}_{x_0}(\cdot)) \lesssim \delta + G \max_{\eta' \in [0, 2\eta]} \mathbb{E}_{X \sim p^* \mathcal{N}(0, \eta' I)} \|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2,$$

where we denote $G = \eta + \frac{\eta^2}{\sigma^2} (d + \log(1/\delta))$.

Proof of Corollary E.6. We write $\mathbb{P}_{\bar{\sigma}}(\cdot)$ to be the probability distribution of $(\bar{x}, V)$ under $\bar{x} \sim \bar{p}$ and $V \sim \mathcal{N}(0, \bar{\sigma}^2 I)$. Fix $\delta \in (0, \frac{1}{2}]$ and write $D = d_* + \log(d/\delta)$, $K = \max\{c^{-1} \log(d^{10}/\delta^2), 1\}$.

Recall that $\tilde{\mathbf{s}}^*(x) = \mathbf{s}^*(x) + \frac{x}{\sigma^2}$ and $\nabla \tilde{\mathbf{s}}^*(x) = \nabla \mathbf{s}^*(x) + \frac{I}{\sigma^2}$. Consider the following set:

$$E = \left\{ (\bar{x}, V) : \text{tr}(\nabla \tilde{\mathbf{s}}^*(\bar{x} + V)) \leq \frac{CD}{\sigma^2}, V^\top \nabla \tilde{\mathbf{s}}^*(\bar{x} + V) V \leq CD \right\},$$

where $C > 0$ is an absolute constant such that by Proposition E.7, it holds that $\mathbb{P}_{\bar{\sigma}}((\bar{x}, V) \notin E) \leq \frac{\delta^2}{d^{10}}$ for $\bar{\sigma} \in [0, 2\sigma]$.

Now, note that for PSD matrix $H \succeq 0$, it holds that $\|H\|_{\text{op}} \leq \|H\|_{\text{F}} \leq \text{tr}(H)$, and $\|HV\|^2 \leq \|H\|_{\text{op}} \cdot V^\top H V$. This immediately implies that with the choice $G = CKD$, it holds that $E \subseteq \mathcal{E}_{G,K}$. This immediately implies that for any $\eta' \in [0, 2\eta]$, it holds that

$$\mathbb{P}_{\bar{x} \sim \bar{p}, V \sim \mathcal{N}(0, (\sigma^2 + \eta') I)}((\bar{x}, V) \notin \mathcal{E}_{G,K}) \leq \frac{\delta^2}{d^{10}}.$$

Applying Proposition E.3 with $\eta \leq \frac{\sigma^2}{4BG}$ gives

$$\mathbb{E}_{x_0 \sim \tilde{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim \delta.$$

Applying Corollary E.6 completes the proof. $\square$

Proof of Theorem 4.7. The proof is the same as the proof of Theorem 4.4, except that we use Corollary E.6 instead. $\square$

The following proposition generalizes the argument of Huang et al. (2024b) that bounds the posterior covariance matrix in terms of the intrinsic dimension. As a remark, we also note that this argument can recover Proposition E.8 (2) easily.

Proposition E.7. Consider the distribution of (X_0, X) under $X_0 \sim \bar{p}$ and $X \sim \mathcal{N}(X_0, \sigma^2 I)$, and we are interested in the following potential function:

$$\mathbb{M}_c(x) := \mathbb{E} \left[\exp \left(c \frac{\|X_0 - D(x)\|^2}{\sigma^2} \right) \middle| X = x \right],$$

where $D(x) = \mathbb{E}[X_0 \mid X = x]$ is the conditional mean.

Then, for any $\bar{\sigma} \geq 0$, we denote $\kappa = \max\{1, \frac{\bar{\sigma}^2}{\sigma^2}\}$, and it holds that

$$\mathbb{E}_{x \sim \bar{p}^* \mathcal{N}(0, \bar{\sigma}^2 I)} \mathbb{M}_{1/(160\kappa)}(x) \leq 4 \inf_{r \geq 0} \exp \left(\log N(\bar{p}; r) + \frac{r^2}{\sigma^2} \right).$$

Proof. We write $q(\cdot | x)$ be the conditional distribution of $X_0 | X = x$ under $X_0 \sim \bar{p}$ and $X \sim \mathcal{N}(X_0, \sigma^2)$, i.e.,

$$q(x_0 | x) = \frac{\exp\left(-\frac{\|\bar{x} + V - X_0\|^2}{2\sigma^2}\right)}{\mathbb{E}_{X_0 \sim \bar{p}}\left[\exp\left(-\frac{\|\bar{x} + V - X_0\|^2}{2\sigma^2}\right)\right]} = \frac{\exp\left(-\frac{\|\bar{x} - X_0\|^2 + 2\langle V, \bar{x} - X_0 \rangle}{2\sigma^2}\right)}{\mathbb{E}_{X_0 \sim \bar{p}}\left[\exp\left(-\frac{\|\bar{x} - X_0\|^2 + 2\langle V, \bar{x} - X_0 \rangle}{2\sigma^2}\right)\right]}.$$

Our goal is to upper bound the moment

$$M(x) := \mathbb{E}_{X_0 \sim q(\cdot | x)} \exp\left(\frac{\|X_0 - D(x)\|^2}{8\sigma^2}\right),$$

where $D(x) := \mathbb{E}_{X_0 \sim q(\cdot | x)}[X_0]$ is the conditional mean. By triangle inequality, we know that for any $\bar{x}$ and

$$M(x) \leq \mathbb{E}_{X_0 \sim q(\cdot | x)} \exp\left(\frac{\|X_0 - \bar{x}\|^2 + \|D(x) - \bar{x}\|^2}{4\sigma^2}\right) \leq \left(\mathbb{E}_{X_0 \sim q(\cdot | x)} \exp\left(\frac{\|X_0 - \bar{x}\|^2}{4\sigma^2}\right)\right)^2. \quad (25)$$

Fix a r -covering of $\bar{\mathcal{X}}$:

$$\bar{\mathcal{X}} \subseteq \bigcup_{i=1}^N B_i,$$

where $N = N(\bar{p}, r)$ and $B_1, \dots, B_N$ are balls of radius r and centers $z_1, \dots, z_N$.

Fix any $\bar{x}$ and an index $j = j(\bar{x})$ such that $\bar{x} \in B_j$.

For any $i \in [N]$, we consider

$$g_i(V) := \mathbb{E}_{X_0 \sim \bar{p}} \mathbb{I}\{X_0 \in B_i\} \exp\left(-\frac{\|\bar{x} - X_0\|^2}{4\sigma^2} + \frac{\langle V, z_i - X_0 \rangle}{\sigma^2}\right).$$

Note that $g_i(V) \equiv 0$ when $i \notin \mathcal{I}$, and also

$$\begin{aligned} \mathbb{E}[g_i(V)] &= \mathbb{E}_{X_0 \sim \bar{p}} \mathbb{I}\{X_0 \in B_i\} \exp\left(-\frac{\|\bar{x} - X_0\|^2}{4\sigma^2} + \frac{\bar{\sigma}^2 \|z_i - X_0\|^2}{2\sigma^4}\right) \\ &\leq \mathbb{E}_{X_0 \sim \bar{p}} \mathbb{I}\{X_0 \in B_i\} \exp\left(\frac{2\kappa r^2 - \|\bar{x} - X_0\|^2}{4\sigma^2}\right) \leq \exp\left(\frac{2\kappa r^2 - (\|z_i - \bar{x}\| - r)_+^2}{4\sigma^2}\right). \end{aligned}$$

In addition, we define (recall $j = j(\bar{x})$ is an index such that $\bar{x} \in B_j$)

$$u(V) := \mathbb{E}_{X_0 \sim \bar{p}} \left[\exp\left(-\frac{\langle V, \bar{x} - X_0 \rangle}{\sigma^2}\right) \middle| X_0 \in B_j \right].$$

Note that

$$\begin{aligned} \mathbb{E}_{X_0 \sim \bar{p}} \left[\exp\left(-\frac{\|\bar{x} - X_0\|^2 + 2\langle V, \bar{x} - X_0 \rangle}{2\sigma^2}\right) \right] &\geq \mathbb{E}_{X_0 \sim \bar{p}} \mathbb{I}\{X_0 \in B_j\} \exp\left(-\frac{\|\bar{x} - X_0\|^2 + 2\langle V, \bar{x} - X_0 \rangle}{2\sigma^2}\right) \\ &\geq \bar{p}(B_j) e^{-2r^2/\sigma^2} u(V), \end{aligned}$$

and we also have

$$\begin{aligned} \mathbb{E}_V [u(V)^{-1}] &\leq \mathbb{E}_V \mathbb{E}_{X_0 \sim \bar{p}} \left[\exp\left(\frac{\langle V, \bar{x} - X_0 \rangle}{\sigma^2}\right) \middle| X_0 \in B_j \right] \\ &= \mathbb{E}_{X_0 \sim \bar{p}} \left[\exp\left(\frac{\bar{\sigma}^2 \|\bar{x} - X_0\|^2}{2\sigma^4}\right) \middle| X_0 \in B_j \right] \leq \exp\left(\frac{2\kappa r^2}{\sigma^2}\right). \end{aligned}$$

By definition and (25), for any V , we can decompose

$$\sqrt{M(\bar{x} + V)} \leq \frac{e^{2r^2/\sigma^2}}{p(B_j)u(V)} \sum_{i \in \mathcal{I}} g_i(V) \exp\left(\frac{\langle V, \bar{x} - z_i \rangle}{\sigma^2}\right).$$

Therefore, by union bound, we know that $\mathbb{P}_V(\mathcal{V}_1 \cap \mathcal{V}_2) \geq 1 - \delta$, where

$$\begin{aligned} \mathcal{V}_1 &= \left\{ u(V)^{-1} \leq e^{2\kappa r^2/\sigma^2} \cdot \frac{3N}{\delta} \right\} \cap \left\{ g_i(V) \leq \exp\left(\frac{2\kappa r^2 - (\|z_i - \bar{x}\| - r)_+^2}{4\sigma^2}\right) \cdot \frac{3N}{\delta} \right\}, \\ \mathcal{V}_2 &= \left\{ \langle V, \bar{x} - z_i \rangle \leq \bar{\sigma} \|\bar{x} - z_i\| \sqrt{2 \log(3N/\delta)} \right\}. \end{aligned}$$

Note that under $V \in \mathcal{V}_1 \cap \mathcal{V}_2$, we can upper bound

$$\begin{aligned} \frac{1}{2} \log M(\bar{x} + V) - \log \frac{1}{p(B_j)} &\leq 2 \log(3N/\delta) + \frac{5\kappa r^2}{\sigma^2} + \max_i -\frac{(\|z_i - \bar{x}\| - r)_+^2}{4\sigma^2} + \sigma^{-1} \|\bar{x} - z_i\| \sqrt{2\kappa \log(3N/\delta)} \\ &\leq 2 \log(3N/\delta) + \frac{5\kappa r^2}{\sigma^2} + \sigma^{-1} r \sqrt{2\kappa \log(3N/\delta)} + 2\kappa \log(3N/\delta) \\ &\leq 5\kappa \log(3N/\delta) + \frac{6\kappa r^2}{\sigma^2}. \end{aligned}$$

Therefore, we denote $w = \frac{1}{10\kappa}$, and we know that

$$\mathbb{P}_V \left(M(\bar{x} + V)^{w/2} \geq \frac{1}{p(B_j)^w} \sqrt{3N/\delta} e^{r^2/\sigma^2} \right) \leq \delta, \quad \forall \delta \in (0, 1).$$

Integration gives

$$\mathbb{E}_V M(\bar{x} + V)^{w/2} \leq \frac{2}{p(B_j)^w} \sqrt{3N} e^{r^2/\sigma^2}.$$

Finally, we can take expectation over $\bar{x} \sim \bar{p}$, and using the fact that $\mathbb{E}[p(B_{j(\bar{x})})^w] \leq \sum_i p(B_i)^{1-w} \leq \sqrt{N}$ gives

$$\mathbb{E}_{\bar{x} \sim \bar{p}, V \sim \mathcal{N}(0, \bar{\sigma}^2 I)} M(\bar{x} + V)^{w/2} \leq 4N e^{r^2/\sigma^2}. \quad (26)$$

This gives the desired upper bound. $\square$

E.5. Technical Lemmas

Proposition E.8. Suppose that $p = \bar{p} * \mathcal{N}(0, \sigma^2 I)$ and $\check{p} = \bar{p} * \mathcal{N}(0, \bar{\sigma}^2 I)$ such that $|\bar{\sigma}^2 - \sigma^2| \leq \frac{\sigma^2}{2}$.

(1) It holds that

$$\mathbb{E}_{X \sim \check{p}} \exp\left(\frac{\sigma^2}{8} \|\nabla \log p(X)\|^2\right) \leq \exp\left(\frac{d}{2}\right).$$

(2) It holds that

$$\mathbb{E}_{X \sim \check{p}} \exp\left(\frac{\sigma^2}{8} \|\nabla^2 \log p(X) + \sigma^{-2} I\|_*\right) \leq \exp\left(\frac{d}{2}\right).$$

(3) Suppose that $|\bar{\sigma}^2 - \sigma^2| \leq \frac{\sigma^2}{8 \max\{\sqrt{d \log(1/\delta)}, \log(1/\delta)\}}$. For any $F : \mathbb{R}^d \rightarrow [0, 1]$, it holds that

$$\mathbb{E}_{X \sim \check{p}} F(X) \leq e \cdot (\mathbb{E}_{X \sim p} F(X) + \delta).$$

Proof. We first prove (3). Denote $\mu = \mathcal{N}(0, \sigma^2 I)$ and $\pi = \mathcal{N}(0, \bar{\sigma}^2 I)$. We denote $\epsilon := \frac{\bar{\sigma}^2}{\sigma^2} - 1$. Then, as long as $\lambda\epsilon < \frac{1}{2}$ and $\epsilon \geq -\frac{1}{2}$,

$$\log(1 + D_{1+\lambda}(\pi \| \mu)) = \frac{d}{2} [-\log(1 - \lambda\epsilon) - \lambda \log(1 + \epsilon)] \leq \frac{d\lambda(1 + \lambda)}{2} \epsilon^2$$

where we use $\log(1 + u) \geq u - u^2$ for $u \geq -\frac{1}{2}$. Then, by the data-processing inequality, this gives

$$1 + D_{1+\lambda}(\check{p} \| p) \leq \exp\left(\frac{d\lambda(1 + \lambda)}{2} \epsilon^2\right), \quad \forall \lambda \leq \frac{1}{2|\epsilon|}.$$

Then, using [Lemma B.7](#) and $\epsilon^2 d \leq \frac{1}{4}$, we know

$$\mathbb{E}_{X \sim \check{p}} F(X) - e \mathbb{E}_{X \sim p} F(X) \leq \min_{1 \leq \lambda \leq \frac{1}{2|\epsilon|}} \exp\left(\frac{d\lambda(1 + \lambda)}{2} \epsilon^2 - \lambda + 1\right) \leq \exp\left(1 - \min\left\{\frac{1}{8d\epsilon^2}, \frac{1}{8|\epsilon|}\right\}\right).$$

This proves (3). Next, we turn toward the proofs of (1) and (2).

We denote $f = \log p$. Then, we can express, for $X_0 \sim \bar{p}$, $X \sim \mathcal{N}(X, \sigma^2 I)$,

$$\nabla f(x) = \frac{1}{\sigma^2} \mathbb{E}[X_0 - X \mid X = x], \quad \nabla^2 f(x) = -\frac{I}{\sigma^2} + \frac{\text{Cov}(X_0 \mid X = x)}{\sigma^4}.$$

For $0 \leq \lambda \leq \frac{\sigma^2}{4}$, we can bound

$$\mathbb{E}_{X \sim p} \exp(\lambda \|\nabla f(X)\|^2) \leq \exp\left(\frac{2d\lambda}{\sigma^2}\right).$$

Therefore, for $\lambda \leq \frac{\sigma^2}{8}$, we have

$$\mathbb{E}_{X \sim \check{p}} \exp(\lambda \|\nabla f(X)\|^2) \leq \sqrt{(1 + D_{\chi^2}(\check{p} \| p)) \mathbb{E}_{X \sim p} \exp(2\lambda \|\nabla f(X)\|^2)} \leq \exp\left(\frac{d}{8} + \frac{2d\lambda}{\sigma^2}\right),$$

where we use $1 + D_{\chi^2}(\check{p} \| p) \leq \exp(\frac{d}{4})$. We can then choose $\lambda = \frac{\sigma^2}{8}$ to complete the proof.

Similarly, for $0 \leq \lambda \leq \frac{\sigma^2}{4}$, we can bound

$$\begin{aligned} \mathbb{E}_{X \sim p} \exp(\lambda \|\nabla^2 f(X) + \sigma^{-2} I\|_*) &\leq \mathbb{E}_{X_0 \sim \bar{p}, X \sim \mathcal{N}(X_0, \sigma^2 I)} \exp\left(\frac{\lambda}{\sigma^4} \|X_0 - X\|^2\right) = \mathbb{E}_{Z \sim \mathcal{N}(0, \sigma^2 I)} \exp\left(\frac{\lambda}{\sigma^4} \|Z\|^2\right) \\ &= (1 - 2\sigma^{-2}\lambda)^{-\frac{d}{2}} \leq \exp\left(\frac{2d\lambda}{\sigma^2}\right). \end{aligned}$$

The upper bound then follows similarly. □

As a direct corollary of [Proposition E.8](#) (1) and (2), we have the following moment bounds.

Lemma E.9. Suppose that [Assumption E.2](#) holds, $0 < \eta \leq \frac{\sigma^2}{4}$, and $0 \leq \lambda \leq \frac{1}{\sigma^2}$. Then the following holds with an absolute constant $C > 0$: For any integer $1 \leq k \leq 16$, and parameter $\eta \in [0, 2\eta]$, it holds that

$$\begin{aligned} (\mathbb{E}_{X \sim p * \mathcal{N}(0, \eta' I)} \|\tilde{\mathbf{s}}^*(X)\|^k)^{1/k} &\leq \frac{C\sqrt{d}}{\sigma}, \\ (\mathbb{E}_{X \sim p * \mathcal{N}(0, \eta' I)} \|\nabla \tilde{\mathbf{s}}^*(X)\|^k)^{1/k} &\leq \frac{Cd}{\sigma^2}. \end{aligned}$$

E.6. Proof of Proposition E.3

We denote

$$W_{z,r,x_0}^*(x) = \sigma^{-2} \langle \dot{\gamma}_{z,r,x_0}(x), u(\gamma_{z,r,x_0}(x)) - u(x_0) \rangle.$$

Then,

$$\log \nu_{x_0}(x) - \log \rho_{x_0}^*(x) = \text{const} + \mathbb{E}_{r,z} \tau_B(W_{z,r,x_0}^*(x)).$$

Therefore,

$$\nabla \log \frac{\nu_{x_0}(x)}{\rho_{x_0}^*(x)} = \mathbb{E}_{r,z} [\nabla W_{z,r,x_0}^*(x) \cdot h_B(W_{z,r,x_0}^*(x))],$$

where

$$h_B(y) = \begin{cases} 1, & y > B, \\ 0, & y \in [-B, B], \\ -1, & y < -B, \end{cases}$$

is the derivative of $\tau_B(y)$. By elementary calculation,

$$\sigma^2 \nabla W_{z,r,x_0}^*(x) = a'_r(u(\gamma_{z,r,x_0}(x)) - u(x_0)) + a_r \nabla u(\gamma_{z,r,x_0}(x)) \cdot \dot{\gamma}_{z,r,x_0}(x).$$

By Lemma B.3 and Lemma B.4, we know that $C_{\text{LSI}}(\rho_{x_0}^*(\cdot)) \leq \eta e^{4B}$. Hence,

$$D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim \eta \mathbb{E}_{x \sim \nu_{x_0}(\cdot)} \left\| \nabla \log \frac{\nu_{x_0}(x)}{\rho_{x_0}^*(x)} \right\|^2.$$

Taking expectation over $x_0 \sim \check{p} = p * \mathcal{N}(0, \eta I)$ gives

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim \eta \mathbb{E}_{z,r,x_0,x} [\|\nabla W_{z,r,x_0}^*(x)\|^2 \cdot \mathbb{I}\{|W_{z,r,x_0}^*(x)| > B\}],$$

where $\mathbb{E}_{z,r,x_0,x}$ is the expectation over $x \sim p$, $x_0 \sim x + \mathcal{N}(0, \eta I)$, $z \sim \mathcal{N}(0, \eta I)$, $r \sim \text{Unif}([0, 1])$.

Bounds on the Moments. Fix any $r \in [0, 1]$. We note that under this distribution,

$$\gamma_{z,r,x_0}(x) \mid x \sim \mathcal{N}(x, \eta((1 - a_r)^2 + b_r^2)I), \quad x_0 \mid x \sim \mathcal{N}(x, \eta I).$$

Therefore, the sub-exponential norm of $\|u(\gamma_{z,r,x_0}(x)) - u(x_0)\|^2$ is upper bounded by $O(d\sigma^2)$, the sub-exponential norm of $\|\nabla u(\gamma_{z,r,x_0}(x))\|_{\text{op}}^2$ is bounded by $O(d/\sigma^2)$, and sub-exponential norm of $\|\dot{\gamma}_{z,r,x_0}(x)\|^2$ is bounded by $O(d\eta)$. Therefore, for a sufficiently large constant C , we can set $M = C(d/\sigma^2 + d^2\eta/\sigma^4) \log^2(1/\delta)$ and bound

$$\mathbb{E}(\|\nabla W_{z,r,x_0}^*(x)\|^2 - M_1)_+ \leq M_1 \delta.$$

Therefore, it holds that

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim M_1(\delta + \mathbb{P}_{z,r,x_0,x}(|W_{z,r,x_0}^*(x)| > B)).$$

Bound on the Tail. Next, we bound $\mathbb{P}_{z,r,x_0,x}(|W_{z,r,x_0}^*(x)| > B)$. We write $W = \dot{\gamma}_{z,r,x_0}(x)$ and $U = \gamma_{z,r,x_0}(x) - x_0$. By definition,

$$W_{z,r,x_0}^*(x) = \langle W, \tilde{\mathbf{s}}^*(x_0 + U) - \tilde{\mathbf{s}}^*(x_0) \rangle = \int_0^1 W^\top \nabla \tilde{\mathbf{s}}^*(x_0 + yU) U dy.$$

Note that

$$\mathbb{I}\{|W_{z,r,x_0}^*(x)| > B\} \leq \frac{2}{B} \tau_{B/2}(W_{z,r,x_0}^*(x)) \leq \frac{2}{B} \int_0^1 \tau_{B/2}(W^\top \nabla \tilde{\mathbf{s}}^*(x_0 + yU) U) dy,$$

where we use the convexity of $y \mapsto \tau_{B/2}(y)$. Hence,

$$\mathbb{P}_{z,r,x_0,x}(|W_{z,r,x_0}^*(x)| > B) \leq \frac{2}{B} \max_{y \in [0,1]} \mathbb{E} \tau_{B/2}(W^\top \nabla \tilde{s}^*(x_0 + yU)U).$$

Note that for any fixed $y \in [0, 1]$,

$$x_0 + yU \mid x \sim \mathbf{N}(x, \eta((1 - ya_r)^2 + y^2 b_r^2)I),$$

and hence the sub-exponential norm of $\|U\|^2$ and $\|W\|^2$ are bounded by $O(d\eta)$, and the sub-exponential norm of $\|\nabla \tilde{s}^*(x_0 + yU)\|_{\text{op}}$ is upper bounded by $O(d/\sigma^2)$. Therefore, we can take $M_2 = C_2 d^2 \eta / \sigma^2 \log^2(1/\delta)$ such that $\mathbb{E}(|W^\top \nabla \tilde{s}^*(x_0 + yU)U| - M_2)_+ \leq M_2 \delta$. To summarize, we have shown that

$$\mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \lesssim \eta M_1 M_2 \cdot (\delta + \max_{y \in [0,1]} \mathbb{P}(|W^\top \nabla \tilde{s}^*(x_0 + yU)U| > \frac{B}{2})).$$

Bound on the Tail Probability. Now, we fix any $y \in [0, 1]$ and consider the joint distribution of $(\bar{x}, x, x_0, z)$ under

$$\bar{x} \sim \bar{p}, \quad x \sim \mathbf{N}(\bar{x}, \sigma^2 I), \quad x_0 \sim \mathbf{N}(x, \eta I), \quad z \sim \mathbf{N}(0, \eta I).$$

We define $V = yU + x_0 - \bar{x}$, and the properties of (U, V, W) are summarized in [Lemma E.10](#).

In the following, we analyze the concentration of (U, W) conditional on $(\bar{x}, V)$. We denote $H := \nabla \tilde{s}^*(x_0 + yU) = \nabla \tilde{s}^*(\bar{x} + V)$. Then, for any $(\bar{x}, V) \in \mathcal{E}_{G,K}$, we can bound $|V^\top H V| + |\text{tr}(H)| \leq \frac{B\sigma^2}{16\eta^2}$, and hence

$$\begin{aligned} \mathbb{P}\left(|W^\top H U| \geq \frac{B}{2} \mid \bar{x}, V\right) &\leq 2 \exp\left(-c_1 \min\left\{\frac{B^2}{\eta^2 \|H\|_{\text{F}}^2 + \eta(\eta/\sigma^2)^2 \|H V\|^2}, \frac{B}{\eta \|H\|_{\text{op}}}\right\}\right) \\ &\leq 2e^{-c_2 K}, \quad \forall (\bar{x}, V) \in \mathcal{E}_{G,K}, \end{aligned}$$

where $c_1, c_2 > 0$ are absolute constants and we use $\eta \leq \frac{\sigma^2}{8BG}$. Hence,

$$\mathbb{P}\left(|W^\top H U| > \frac{B}{2}\right) \leq \mathbb{P}_{\bar{x} \sim \bar{p}, V \sim \mathbf{N}(0, \sigma_V^2 I)}(\mathcal{E}_{G,K}) + 2 \exp(-c_2 K).$$

This is the desired upper bound. □

Lemma E.10. Fix $r, y \in [0, 1]$. Consider the joint distribution of $(\bar{x}, x, x_0, z)$ under

$$\bar{x} \sim \bar{p}, \quad x \sim \mathbf{N}(\bar{x}, \sigma^2 I), \quad x_0 \sim \mathbf{N}(x, \eta I), \quad z \sim \mathbf{N}(0, \eta I).$$

Under this law, we consider the random variables

$$W = \dot{\gamma}_{z,r,x_0}(x), \quad U = \gamma_{z,r,x_0}(x) - x_0, \quad V = y\gamma_{z,r,x_0}(x) + (1 - y)x_0 - \bar{x}.$$

Then, the following holds.

(1) (U, V, W) are jointly Gaussian and independent of $\bar{x}$, and marginally $V \sim \mathbf{N}(0, \sigma_V^2 I)$ with $\sigma_V^2 \in [\sigma^2, \sigma^2 + 2\eta]$.

(2) The conditional distribution of $[U; W] \mid V$ can be written as

$$[U; W] \mid V \sim \mathbf{N}([c_U V, c_W V], \bar{\Sigma}), \quad \bar{\Sigma} = \begin{bmatrix} \bar{\sigma}_U^2 I & \bar{\sigma}_{UW} I \\ \bar{\sigma}_{UW} I & \bar{\sigma}_W^2 I \end{bmatrix},$$

so that $\|\bar{\Sigma}\|_{\text{op}} \leq 3\eta$ and

$$|c_U| \leq \frac{\eta}{\sigma^2}, \quad |c_W| \leq \frac{2\eta}{\sigma^2}, \quad \bar{\sigma}_U^2 \leq \eta, \quad \bar{\sigma}_W^2 \leq 3\eta, \quad |\bar{\sigma}_{UW}| \leq \frac{2\eta^2}{\sigma^2}.$$

(3) Suppose that the symmetric matrix $H = \mathcal{H}(\bar{x} + V)$ is a function of $\bar{x} + V$. Then it holds that

$$\begin{aligned} & \mathbb{P}\left(|W^\top HU| \geq t + \frac{4\eta^2}{\sigma^2}(|V^\top HV| + |\text{tr}(H)|) \mid \bar{x}, V\right) \\ & \leq 2 \exp\left(-c \min\left\{\frac{t^2}{\eta^2\|H\|_F^2 + \eta(\eta/\sigma^2)^2\|HV\|^2}, \frac{t}{\eta\|H\|_{\text{op}}}\right\}\right), \quad \forall t \geq 0, \end{aligned}$$

where $c > 0$ is an absolute constant.

Proof. By definition, we can consider $w_0 := x - \bar{x}$, $w_1 := x_0 - x$, and then

$$w_0 \sim \mathcal{N}(0, \sigma^2 I), \quad w_1 \sim \mathcal{N}(0, \eta I), \quad z \sim \mathcal{N}(0, \eta I),$$

where w_0, w_1, z are independent Gaussian vectors, and (w_0, w_1, z) is independent of $\bar{x} \sim \bar{p}$. Then, we can rewrite

$$W = -a'_r w_1 + b'_r z, \quad U = -a_r w_1 + b_r z, \quad V = w_0 + (1 - y a_r) w_1 + y b_r z.$$

Hence, under this distribution, (U, V, W) are jointly Gaussian and independent of $\bar{x} \sim \bar{p}$. The key fact is that $a_r a'_r + b_r b'_r = 0$ (because $a_r^2 + b_r^2 \equiv 1$ for $r \in [0, 1]$), and hence $\mathbb{E}[UW^\top] = 0$. Therefore, the joint distribution of $[V; U; W]$ can be written

$$\begin{bmatrix} V \\ U \\ W \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_V^2 I & \sigma_{VU} I & \sigma_{VW} I \\ \sigma_{VU} I & \sigma_U^2 I & 0 \\ \sigma_{VW} I & 0 & \sigma_W^2 I \end{bmatrix}\right),$$

and we can calculate

$$\begin{aligned} \sigma_V^2 &= \sigma^2 + \eta(1 - y a_r)^2 + \eta y^2 b_r^2, & \sigma_U^2 &= \eta, & \sigma_W^2 &= (\pi/2)^2 \eta, \\ \sigma_{VU} &= \eta(y b_r^2 - a_r(1 - y a_r)), & \sigma_{VW} &= \eta(y b_r b'_r - a'_r(1 - y a_r)). \end{aligned}$$

In particular, we know $\sigma_V^2 \in [\sigma^2, \sigma^2 + 2\eta]$, $|\sigma_{VU}| \leq \eta$, $|\sigma_{VW}| \leq 2\eta$, and $\sigma_W^2 \leq 3\eta$. Further, we can write down

$$\begin{aligned} c_U &= \frac{\sigma_{VU}}{\sigma_V^2}, & c_W &= \frac{\sigma_{VW}}{\sigma_V^2}, \\ \bar{\sigma}_U^2 &= \sigma_U^2 - \frac{\sigma_{VU}^2}{\sigma_V^2}, & \bar{\sigma}_W^2 &= \sigma_W^2 - \frac{\sigma_{VW}^2}{\sigma_V^2}, & \bar{\sigma}_{UW} &= -\frac{\sigma_{VU}\sigma_{VW}}{\sigma_V^2}. \end{aligned}$$

The desired statement (2) then follows.

Finally, to prove (3), we condition on $(V, \bar{x})$ and apply Gaussian concentration (Lemma B.5) to the matrix $A = \begin{bmatrix} 0 & H \\ H & 0 \end{bmatrix}$ and the Gaussian vector $[U; W] \mid (V, \bar{x}) \sim \mathcal{N}(v, \bar{\Sigma})$, where $v = [c_U V, c_W V]$. Note that $v^\top A v = 2c_U c_W V^\top H V$, $\text{tr}(\bar{\Sigma} A) = 2\bar{\sigma}_{UW} \text{tr}(H)$, $\|A v\|^2 = (c_U^2 + c_W^2)\|H V\|^2$, and $\|A\|_{\text{op}} \leq \|H\|_{\text{op}}$. Therefore, by Lemma B.5, for $t \geq 0$,

$$\begin{aligned} & \mathbb{P}(|W^\top HU - 2(c_U c_W V^\top H V^\top + \bar{\sigma}_{UW} \text{tr}(H))| > t \mid \bar{x}, V) \\ & \leq 2 \exp\left(-\frac{1}{16} \min\left\{\frac{t^2}{\eta^2\|H\|_F^2 + \eta(c_U^2 + c_W^2)\|H V\|^2}, \frac{t}{\eta\|H\|_{\text{op}}}\right\}\right). \end{aligned}$$

The desired upper bound then follows from bounding $|c_U| \leq \frac{\eta}{\sigma^2}$, $|c_W| \leq \frac{2\eta}{\sigma^2}$, and $\bar{\sigma}_{UW} \leq \frac{2\eta^2}{\sigma^2}$. $\square$

E.7. Proof of Proposition E.4

We denote

$$W_{z,r,x_0}^*(x) := \langle \dot{\gamma}_{z,r,x_0}(x), \tilde{\mathbf{s}}^*(\gamma_{z,r,x_0}(x)) - \tilde{\mathbf{s}}^*(x_0) \rangle, \quad W_{z,r,x_0}(x) := \langle \dot{\gamma}_{z,r,x_0}(x), \tilde{\mathbf{s}}(\gamma_{z,r,x_0}(x)) - \tilde{\mathbf{s}}(x_0) \rangle. \quad (27)$$

Then, we can abbreviate

$$\begin{aligned} \log \hat{\rho}_{x_0}(x) &= \text{const} + \mathbb{E}_{r,z} \text{Clip}_B(W_{z,r,x_0}(x)) - \frac{\|x - x_0 - \bar{\eta} \mathbf{s}(x_0)\|^2}{2\bar{\eta}}, \\ \log \rho_{x_0}^*(x) &= \text{const} + \mathbb{E}_{r,z} \text{Clip}_B(W_{z,r,x_0}^*(x)) - \frac{\|x - x_0 - \bar{\eta} \mathbf{s}^*(x_0)\|^2}{2\bar{\eta}}. \end{aligned}$$

Decomposition of KL Divergence. As an intermediate distribution, we consider the density $\tilde{\rho}_{x_0}(\cdot)$ with

$$\log \tilde{\rho}_{x_0}(x) = \text{const} + \mathbb{E}_{r,z} \text{Clip}_B(W_{z,r,x_0}(x)) - \frac{\|x - x_0 - \bar{\eta} \mathbf{s}^*(x_0)\|^2}{2\bar{\eta}}.$$

Note that for $u, v \in \mathbb{R}^d$, it holds that

$$\log(1 + D_{\chi^2}(\mathbf{N}(u, \bar{\eta}I) \parallel \mathbf{N}(v, \bar{\eta}I))) = \frac{\|u - v\|^2}{\bar{\eta}}.$$

Therefore, using [Lemma B.8](#), [Lemma B.9](#), and the fact that $\log(1 + Ct) \leq C \log(1 + t)$ for $t \geq 0$ and $C \geq 1$, we know that for any x_0 ,

$$D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \hat{\rho}_{x_0}(\cdot)) \leq 2D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \tilde{\rho}_{x_0}(\cdot)) + \bar{\eta}e^{4B} \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2.$$

Next, we note that we can write $\tilde{\rho}_{x_0}(\cdot) \propto \rho_{x_0}^*(\cdot) e^{h_{x_0}}$, where

$$h_{x_0}(x) = \mathbb{E}_{r,z} \text{Clip}_B(W_{z,r,x_0}(x)) - \mathbb{E}_{r,z} \text{Clip}_B(W_{z,r,x_0}^*(x)) \in [-2B, 2B].$$

Then, using [Lemma B.11](#), we can bound

$$D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \tilde{\rho}_{x_0}(\cdot)) \leq 4e^{2B} (D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + \mathbb{E}_{x \sim \nu_{x_0}(\cdot)} [h_{x_0}(x)^2]).$$

Combining the inequalities above immediately implies

$$\begin{aligned} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \hat{\rho}_{x_0}(\cdot)) &\leq 8e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \\ &\quad + \bar{\eta}e^{4B} \mathbb{E}_{x_0 \sim \check{p}} \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2 + 8e^{2B} \mathbb{E}_{x_0 \sim \check{p}, x \sim \nu_{x_0}(\cdot)} [h_{x_0}(x)^2]. \end{aligned} \quad (28)$$

Further, using $D_{\text{H}}^2(\mu_1, \mu_2) \leq \min\{\frac{1}{2} D_{\text{KL}}(\mu_1 \parallel \mu_2), 1\}$, we can upper bound

$$\begin{aligned} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{H}}^2(\nu_{x_0}(\cdot), \hat{\rho}_{x_0}(\cdot)) &\leq 8e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) \\ &\quad + e^{4B} \mathbb{E}_{x_0 \sim \check{p}} \min\{\bar{\eta} \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2, 1\} + 8e^{2B} \mathbb{E}_{x_0 \sim \check{p}, x \sim \nu_{x_0}(\cdot)} [h_{x_0}(x)^2]. \end{aligned} \quad (29)$$

Upper Bound on h . We denote

$$\phi_{\eta'} := \mathbb{E}_{X \sim p * \mathbf{N}(0, \eta'I)} \min\{G \|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2, B^2\}, \quad \bar{\phi} := \max_{\eta' \in [0, 2\eta]} \phi_{\eta'}. \quad (30)$$

In the following, we will show that $\mathbb{E}_{x_0 \sim \check{p}, x \sim \nu_{x_0}(\cdot)} [h_{x_0}(x)^2]$ can be upper bounded by $\bar{\phi}$.

We upper bound

$$\begin{aligned} \mathbb{E}_{x_0 \sim \check{p}, x \sim \nu_{x_0}(\cdot)} [h_{x_0}(x)^2] &= \mathbb{E}_{x \sim p, x_0 \sim \mathbf{N}(x, \eta'I)} [h_{x_0}(x)^2] \\ &\leq \max_{r \in [0, 1]} \mathbb{E}_{x, x_0, z} \text{Clip}_{2B}(W_{z,r,x_0}(x) - W_{z,r,x_0}^*(x))^2 \\ &\lesssim \max_{r \in [0, 1]} \mathbb{E}_{x, x_0, z} \text{Clip}_{2B}(\langle \dot{\gamma}_{z,r,x_0}(x), \mathbf{s}^*(\gamma_{z,r,x_0}(x)) - \mathbf{s}(\gamma_{z,r,x_0}(x)) \rangle)^2 \\ &\quad + \max_{r \in [0, 1]} \mathbb{E}_{x, x_0, z} \text{Clip}_{2B}(\langle \dot{\gamma}_{z,r,x_0}(x), \mathbf{s}^*(x_0) - \mathbf{s}(x_0) \rangle)^2, \end{aligned}$$

where the expectation $\mathbb{E}_{x, x_0, z}[\cdot]$ is taken over $x \sim p$, $x_0 \sim x + \mathbf{N}(0, \eta'I)$, $z \sim \mathbf{N}(0, \eta'I)$.

We fix $r \in [0, 1]$ and first analyze $W_r := \dot{\gamma}_{z,r,x_0}(x)$, $U_r = \gamma_{z,r,x_0}(x) - x$. Note that (U_r, W_r) is jointly Gaussian and independent of $x \sim p$, and we can write down the distribution of $[U_r; W_r]$ as $[U_r; W_r] \sim \mathbf{N}\left(0, \begin{bmatrix} \sigma_U^2 I & \sigma_{UW} I \\ \sigma_{UW} I & \sigma_W^2 I \end{bmatrix}\right)$ such that

$$\sigma_U^2 = \eta((1 - a_r)^2 + b_r^2), \quad \sigma_W^2 = (\pi/2)^2 \eta, \quad \sigma_{UW} = -\eta a'_r.$$

Applying [Lemma E.11](#) to the function $g(x) = \mathbf{s}^*(x) - \mathbf{s}(x)$ gives

$$\begin{aligned} \mathbb{E} \text{Clip}_{2B}(\langle \dot{\gamma}_{z,r,x_0}(x), \mathbf{s}^*(\gamma_{z,r,x_0}(x)) - \mathbf{s}(\gamma_{z,r,x_0}(x)) \rangle) &= \mathbb{E} \text{Clip}_{2B}(\langle W_r, \mathbf{s}^*(x + U_r) - \mathbf{s}(x + U_r) \rangle)^2 \\ &\lesssim B^2 \delta + \mathbb{E} \min\{G \|\mathbf{s}^*(\gamma_{z,r,x_0}(x)) - \mathbf{s}(\gamma_{z,r,x_0}(x))\|^2, B^2\} \\ &\leq B^2 \delta + \bar{\phi}, \end{aligned}$$

where we recall that we denote $G = \eta + \frac{\eta^2}{\sigma^2}(d + \log(1/\delta))$. Similarly, we can bound

$$\mathbb{E} \text{Clip}_{2B}(\langle \dot{\gamma}_{z,r,x_0}(x), \mathbf{s}^*(x_0) - \mathbf{s}(x_0) \rangle)^2 \lesssim B^2 \delta + \mathbb{E} \min\{G \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2, B^2\} \leq B^2 \delta + \bar{\phi}.$$

Combining these inequalities, we can conclude that

$$\mathbb{E}_{x_0 \sim q, x \sim \nu_{x_0}(\cdot)}[h_{x_0}(x)^2] \lesssim B^2 \delta + \bar{\phi}, \quad (31)$$

where $\bar{\phi}$ is defined in [Eq. \(30\)](#).

Putting Everything Together. We first prove [Proposition E.4](#) (1). Combining [Eq. \(28\)](#), [Eq. \(31\)](#) and the definition of $\phi_{\eta'}$ in [Eq. \(30\)](#), we get

$$\begin{aligned} &\mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \hat{\rho}_{x_0}(\cdot)) \\ &\lesssim e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + \bar{\eta} e^{4B} \mathbb{E}_{x_0 \sim \check{p}} \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2 + e^{2B} \left(B^2 \delta + \max_{\eta' \in [0, 2\eta]} \phi_{\eta'} \right) \\ &\lesssim e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + e^{2B} B^2 \delta + e^{4B} G \max_{\eta' \in [0, 2\eta]} \mathbb{E}_{X \sim p^* \mathbf{N}(0, \eta' I)} \|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2, \end{aligned}$$

where we use $\bar{\eta} \leq \eta \leq G$.

Similarly, combining [Eq. \(29\)](#), [Eq. \(31\)](#), we can bound

$$\begin{aligned} &\mathbb{E}_{x_0 \sim \check{p}} D_{\text{H}}^2(\nu_{x_0}(\cdot), \hat{\rho}_{x_0}(\cdot)) \\ &\lesssim e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + e^{4B} \mathbb{E}_{x_0 \sim \check{p}} \min\{\bar{\eta} \|\mathbf{s}^*(x_0) - \mathbf{s}(x_0)\|^2, 1\} + e^{2B} \left(B^2 \delta + \max_{\eta' \in [0, 2\eta]} \phi_{\eta'} \right) \\ &\lesssim e^{2B} \mathbb{E}_{x_0 \sim \check{p}} D_{\text{KL}}(\nu_{x_0}(\cdot) \parallel \rho_{x_0}^*(\cdot)) + e^{2B} B^2 \delta + e^{4B} \bar{B}^2 \max_{\eta' \in [0, 2\eta]} \mathbb{E}_{X \sim p^* \mathbf{N}(0, \eta' I)} \min\{\|\mathbf{s}(X) - \mathbf{s}^*(X)\|^2, 1\}. \end{aligned}$$

Combining the inequality above with [Proposition E.8](#) (3) completes the proof of [Proposition E.4](#) (2). $\square$

Lemma E.11. Suppose that $p = \bar{p} * \mathbf{N}(0, \sigma^2 I)$, and consider $x \sim p$, $(U, W) \sim \mathbf{N}(0, \Sigma)$ is independent of x , with $\Sigma = \begin{bmatrix} \sigma_U^2 I & \sigma_{UW} I \\ \sigma_{UW} I & \sigma_W^2 I \end{bmatrix}$. Then, it holds that for any $\delta \in (0, 1)$,

$$\mathbb{E} \text{Clip}_B(\langle W, g(x + U) \rangle)^2 \leq 2B^2 \delta + 4 \mathbb{E} \min\left\{ \left(\sigma_W^2 + \frac{3\sigma_{UW}^2}{\sigma^2}(d + \log(1/\delta)) \right) \|g(x + U)\|^2, B^2 \right\}.$$

Proof. We consider the joint distribution of

$$\bar{x} \sim \bar{p}, \quad x \sim \mathbf{N}(\bar{x}, \sigma^2 I), \quad [U; W] \sim \mathbf{N}(0, \Sigma).$$

We write $V = x - \bar{x} + U$. Then, (V, W) is independent of $\bar{x} \sim \bar{p}$ and the conditional distribution $W \mid V$ can be written as

$$W \mid V \sim \mathbf{N}(cV, \bar{\sigma}^2 I),$$

where $c = \frac{\sigma_{VW}}{\sigma^2 + \sigma_U^2}$, $\bar{\sigma}^2 = \sigma_W^2 - \frac{\sigma_{VW}^2}{\sigma^2 + \sigma_U^2}$.

Now, note that for random vector X and a fixed vector v , we can bound

$$\begin{aligned} \mathbb{E} \text{Clip}_B(\langle X, v \rangle)^2 &= \mathbb{E} \min\{\langle X, v \rangle^2, B^2\} \\ &\leq \min\{\mathbb{E} \langle X, v \rangle^2, B^2\} = \min\{v^\top (\mathbb{E}[X X^\top]) v, B^2\}. \end{aligned}$$

Then, we can bound

$$\begin{aligned}\mathbb{E} \text{Clip}_B(\langle W - cV, g(\bar{x} + V) \rangle)^2 &= \mathbb{E}[\mathbb{E}[\text{Clip}_B(\langle W - cV, g(\bar{x} + V) \rangle)^2 \mid \bar{x}, V]] \\ &\leq \mathbb{E}[\min\{g(\bar{x} + V)^\top \mathbb{E}[(W - cV)(W - cV)^\top \mid \bar{x}, V]g(\bar{x} + V), B^2\}] \\ &= \mathbb{E} \min\{\bar{\sigma}^2 \|g(\bar{x} + V)\|^2, B^2\}.\end{aligned}$$

Further, by [Lemma B.5](#), we know that $\mathbb{P}(\|V\|^2 \geq (\sigma^2 + \sigma_U^2)(2d + 3 \log(1/\delta))) \leq \delta$. This implies

$$\mathbb{E} \text{Clip}_B(\langle cV, g(\bar{x} + V) \rangle)^2 \leq B^2 \delta + \mathbb{E} \min\{c^2(\sigma^2 + \sigma_U^2)(2d + 3 \log(1/\delta))\|g(\bar{x} + V)\|^2, B^2\}.$$

Combining the inequalities, we can conclude

$$\begin{aligned}\mathbb{E} \text{Clip}_B(\langle W, g(x + U) \rangle)^2 &= \mathbb{E} \text{Clip}_B(\langle W, g(\bar{x} + V) \rangle)^2 \\ &\leq 2 \mathbb{E} \text{Clip}_B(\langle cV, g(\bar{x} + V) \rangle)^2 + 2 \mathbb{E} \text{Clip}_B(\langle W - cV, g(\bar{x} + V) \rangle)^2 \\ &\leq 2B^2 \delta + 4 \mathbb{E} \min\left\{\left(\sigma_W^2 + \frac{3\sigma_{UV}^2}{\sigma^2}(d + \log(1/\delta))\right)\|g(\bar{x} + V)\|^2, B^2\right\}.\end{aligned}$$

This gives the desired upper bound. $\square$

F. Log-Concave Sampling

Here, we apply [Theorem 3.3](#) at each step of [Algorithm 3](#) to sample from the RGO distribution. Note that [Theorem 3.3](#) requires the choice of a point x_+ which is close to $\text{prox}_{\eta f}(x_0)$. For simplicity, we assume that this error is zero, which amounts to assuming that we have access to the proximal oracle for f . When f is β -smooth and $h \leq 1/(2\beta)$, then computation of the proximal map is a convex optimization problem, and error analysis can be done as in [Altschuler & Chewi \(2024\)](#). Otherwise, one could generalize the analysis to consider obtaining a stationary point of the proximal map optimization as in [Fan et al. \(2023\)](#).

We recall the definitions of PI and LSI in [Definition B.1](#) and [Definition B.2](#).

Theorem F.1. *Let $\lambda \geq 2$ be a fixed constant, and $\varepsilon \in (0, \frac{1}{2}]$. Suppose that $\mu \propto e^{-f}$ and that f satisfies [Assumption 3.2](#). Choose*

$$\eta^{-1} = C \left(\beta_s^2 d^s \log(1/\varepsilon) + \beta_s^2 d^{-(1-s)} \log^2(1/\varepsilon) \right)^{1/(1+s)}$$

for a sufficiently large universal constant $C = C_\lambda > 0$. Let $\hat{\mu}$ denote the law of the output of [Algorithm 3](#) initialized at μ_0 , where in each step the RGO is implemented by [Algorithm 1](#) via [Theorem 3.3](#). Then, the following holds.

1. Suppose that μ satisfies a log-Sobolev inequality with constant $C_{\text{LSI}}(\mu) < \infty$, which can only hold if $s = 1$ (i.e., f is smooth). Then, $\mathcal{R}_\lambda(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ using at most

$$N = \tilde{O}\left(\kappa d^{1/2} \log^{3/2} \frac{\mathcal{R}_\lambda(\mu_0 \parallel \mu)}{\varepsilon^2} + \kappa \log^2 \frac{\mathcal{R}_\lambda(\mu_0 \parallel \mu)}{\varepsilon^2}\right) \quad \text{first-order queries in expectation,}$$

where $\kappa := C_{\text{LSI}}(\mu) \beta_1$ is the condition number.

2. Suppose that μ satisfies a Poincaré inequality with constant $C_{\text{PI}}(\mu) < \infty$. Then, $D_{\chi^2}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ using at most

$$N = \tilde{O}\left(C_{\text{PI}}(\mu) \beta_s^{2/(1+s)} d^{s/(1+s)} \log^{1/(1+s)}\left(\frac{1}{\varepsilon}\right) \left(1 + \frac{\log^{1/(1+s)}(1/\varepsilon)}{d^{(1-s)/s}}\right) \log \frac{D_{\chi^2}(\mu_0 \parallel \mu)}{\varepsilon^2}\right)$$

first-order queries in expectation.

3. Suppose that π is log-concave. Then, $D_{\text{KL}}(\hat{\mu} \parallel \mu) \leq \varepsilon^2$ using at most

$$N = \tilde{O}\left(\beta_s^{2/(1+s)} d^{s/(1+s)} \frac{W_2^2(\mu_0, \mu)}{\varepsilon^2}\right) \quad \text{first-order queries in expectation.}$$

Proof. This follows from tracking the error of the RGO implementation and choosing δ appropriately, exactly as in [Altschuler & Chewi \(2024\)](#), and is therefore omitted. $\square$

This theorem can be generalized in several directions. One could assume that π satisfies a Latała–Oleszkiewicz inequality which interpolates between Poincaré and log-Sobolev, as in [Chen et al. \(2022\)](#); [Chewi et al. \(2024\)](#). One could also consider RGO implementations under more complicated settings, such as composite settings ([Fan et al., 2023](#)). For brevity, we omit such extensions.